Information Fusion 119 (2025) 103027

Contents lists available at ScienceDirect

Information Fusion

journal homepage: www.elsevier.com/locate/inffus

From patches to WSIs: A systematic review of deep Multiple Instance Learning in computational pathology

Yuchen Zhang [a,1], Zeyu Gao [b,c,1], Kai He [d], Chen Li [a,*], Rui Mao [e]

[a] School of Computer Science and Technology, Xi'an Jiaotong University, 28 Xianning West Road, Xi'an, 710049, Shaanxi, China
[b] Department of Oncology, University of Cambridge, Cambridge, UK
[c] CRUK Cambridge Centre, University of Cambridge, Cambridge, UK
[d] National University of Singapore, Singapore
[e] Nanyang Technological University, Singapore

ARTICLE INFO

Keywords:
Multiple Instance Learning
Computational pathology
Whole slide images
Self-supervised learning
Multimodal fusion

ABSTRACT

Clinical decision support systems for pathology, particularly those utilizing computational pathology (CPATH) for whole slide image (WSI) analysis, face significant challenges due to the need for high-quality annotated datasets. Given the vast amount of information contained in WSIs, creating such datasets is often prohibitively expensive and time-consuming. Multiple Instance Learning (MIL) has emerged as a promising alternative, enabling training that relies solely on coarse-grained supervision by the fusion of extensive localized information from large-scale wholes, thereby reducing the dependency on costly pixel-level labeling. As a result, MIL has become a pivotal technique in CPATH, driving a surge in related research, particularly over the past five years. This expanding body of work has catalyzed technological innovation, introduced transformative advancements in the field, and been further accelerated by progress in deep learning architectures, large-scale pretraining strategies, and Large Language Models (LLMs). This paper provides a systematic review of recent developments in deep MIL methods, analyzing technological advancements from multiple perspectives, including encoder backbone architectures, encoder pretraining strategies, and MIL aggregation techniques. We present a comprehensive overview of progress in each domain, catalog specific application scenarios, and highlight pivotal contributions that have shaped the field. Finally, we explore emerging research directions and potential future challenges for MIL-based CPATH.

1. Introduction

In clinical practice, pathological evaluation of biopsy and surgical slides is essential for decision-making but is time-consuming and prone to subjectivity, leading to inconsistent diagnoses. The shortage of qualified pathologists [1] highlights the need for automated clinical tools. While advancements in scanner technology [2,3] have improved the quality and availability of Whole Slide Images (WSIs), digitization alone does not significantly enhance diagnostic workflows, primarily aiding teaching and remote consultations [4,5]. Computational pathology (CPATH) addresses these limitations by using advanced algorithms to analyze WSIs, improving diagnostic accuracy [6–8].

Fully supervised CPATH methods have made strides in classification, segmentation, and regression tasks using machine learning, particularly deep learning [9–12]. However, CPATH faces challenges not encountered in fields like intelligent security or autonomous driving, where transfer learning and crowdsourced annotations are common [13]. First, pathological images differ significantly from natural images like those in ImageNet [14], creating a domain gap that hampers transfer learning. Second, WSIs are high-resolution and multi-scale, requiring expert annotations—contradicting the goal of reducing pathologists' workload [15–17]. For example, the widely used CAMELYON16 dataset contains only 400 finely annotated WSIs [18], underscoring the challenge of acquiring large-scale annotated data. To address these issues, CPATH models must leverage coarse-grained annotations, such as WSI-level labels from pathology reports, enabling the creation of large-scale datasets [19,20].

Multiple Instance Learning (MIL) [21,22] has emerged as a promising approach for CPATH, enabling various tasks without the need for fine-grained annotations. Unlike its original formulation in conventional machine learning, where MIL aims to classify individual

* Correspondence to: School of Computer Science and Technology, Xi'an Jiaotong University, Xi'an, Shaanxi, China.
E-mail address: cli@xjtu.edu.cn (C. Li).
[1] These authors contributed equally.

https://doi.org/10.1016/j.inffus.2025.103027
Received 26 December 2024; Received in revised form 7 February 2025; Accepted 8 February 2025
Available online 18 February 2025
1566-2535/© 2025 Published by Elsevier B.V.

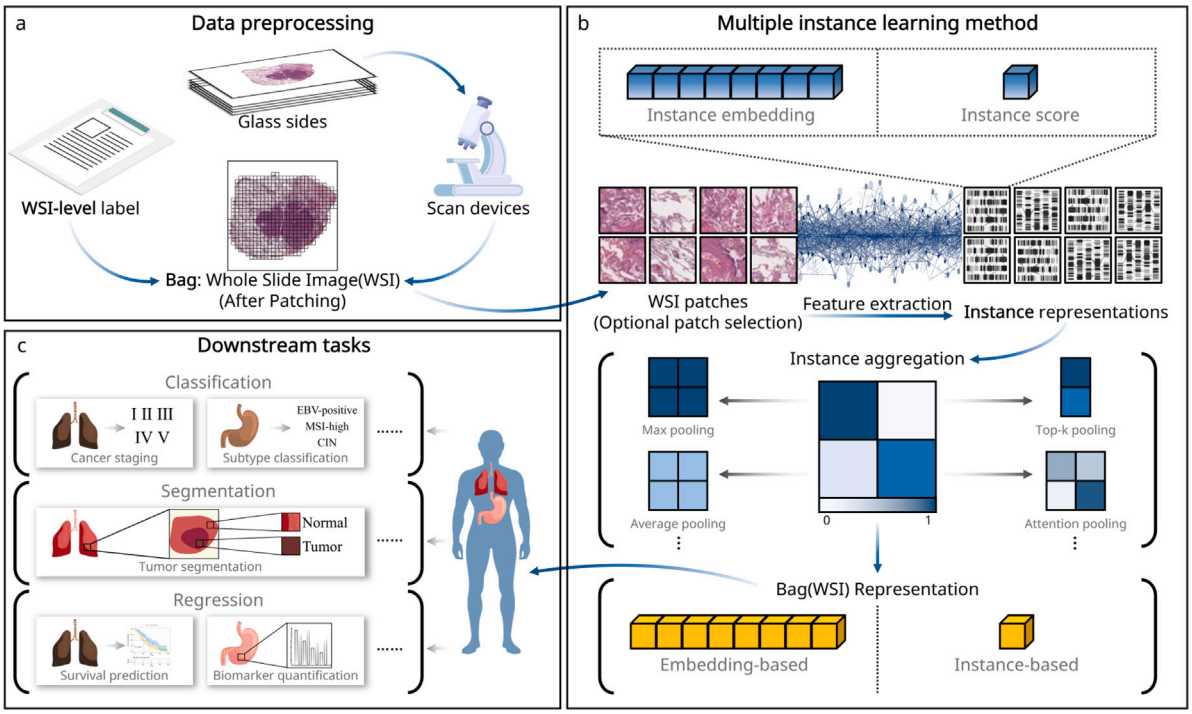

Fig. 1. Deep MIL workflow in CPATH.

instances within a bag using bag-level supervision, MIL in CPATH primarily focuses on classifying entire bags, i.e., WSIs. In this context, a WSI is divided into patches, which are processed using WSI-level annotations for feature extraction and aggregation. This allows tasks such as classification, outcome regression, and segmentation. For instance, a negative WSI label indicates the absence of tumor-containing patches, while a positive label implies the presence of at least one such patch, exemplifying the standard multiple instance (SMI) assumption [21]. By disassembling the WSI into patches and aggregating features at the instance level, MIL can extract high-quality, fine-grained features with only coarse-grained supervision. Compared to traditional methods, MIL reduces the need for fine-grained annotations, enabling predictions at the WSI or even patch level based on readily available pathology reports. This facilitates the creation of large-scale pathology datasets, accelerating the development of robust CPATH models. The field has witnessed substantial growth over the past two decades, with substantial advancements in deep learning architectures such as Vision Transformers [23], DEtection TRansformers [24], self-supervised learning (SSL) [25,26], and Large Language Models (LLMs) [27]. These innovations necessitate an updated review to summarize recent progress and explore future directions.

To provide a comprehensive overview, we systematically searched Google Scholar using the terms "pathology", "multiple instance learning", and "deep learning". After an initial screening, 190 relevant articles published between January 2018 and July 2024 were selected. This review focuses on MIL innovations in CPATH, excluding review articles, non-deep learning algorithms (*e.g.*, SVM), and purely dataset construction efforts. We prioritized impactful papers published in top journals and conferences, including *Nature*, *Nature Medicine*, *MedIA* (Medical Image Analysis), *TMI* (IEEE Transactions on Medical Imaging), and *CVPR* (The IEEE/CVF Computer Vision and Pattern Recognition Conference).

Despite recent efforts to summarize advancements in MIL, many works either provide brief overviews with insufficient coverage [28,29] or lack a clear technological road-map [30–32], leaving readers unable to fully grasp the logical progression of MIL techniques in CPATH. A detailed and systematic review is therefore essential, encompassing fundamental concepts, recent developments, and future directions to guide researchers and practitioners in the field. This paper systematically reviews the past six years of MIL advancements in CPATH, organizing and analyzing a substantial body of studies. The review focuses on three aspects: (1) the foundational frameworks for applying MIL in CPATH, offering insight into its core principles; (2) the evolution of MIL components over time, identifying major trends and turning points; and (3) the two critical elements of MIL frameworks—MIL architectures and feature representation—detailing key studies, technical advancements, and their interrelations to provide a comprehensive understanding of the field. The structure of the remaining sections is as follows: Section 2 introduces the fundamental principles of deep MIL in CPATH. Section 3 examines the key advancements of MIL in CPATH over the past six years, with a particular focus on the significant changes around 2021, and discusses the technological trends and transformations. Section 4 focuses on MIL architectures and feature representation, highlighting pivotal works and tracing the technical development trajectories in CPATH. Section 5 discusses future directions from both technical and application perspectives. Finally, Section 6 summarizes the contributions and key findings.

2. Foundation of MIL in CPATH

MIL is a form of weakly supervised learning typically utilized when training data can be decomposed into hierarchical structures adhering to the general SMI assumption. In CPATH, MIL methods generally follow a two-step pipeline: feature extraction and aggregation, with task-specific heads applied to address particular objectives. Unlike conventional MIL, which treats instances uniformly, MIL in CPATH is designed to address the unique uncertainties inherent in pathology data, such as variability in staining and tumor heterogeneity—factors that can substantially influence diagnostic accuracy. By focusing on extracting salient features from WSIs, MIL in CPATH captures these complexities effectively. The MIL pipeline in CPATH is illustrated in Fig. 1.

(a) Data preprocessing. Consider a dataset of B WSIs, denoted as $\{(X_i^{raw}, Y_i) \mid i = 1, \ldots, B\}$, which are digitized using scanning devices

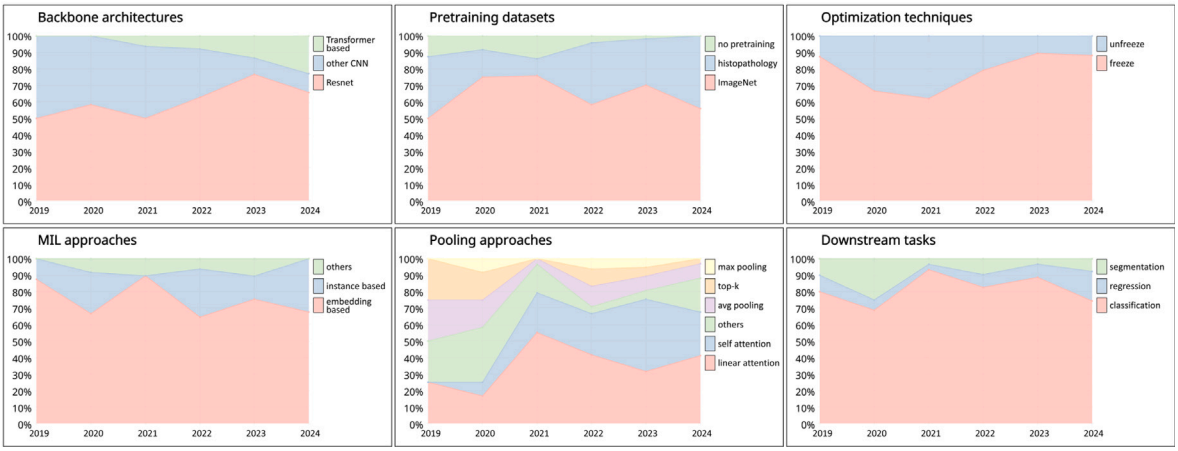

Fig. 2. The development trends of MIL in CPATH in the past six years.

and contain billions of pixels. The corresponding WSI-level annotations, $\mathbf{Y} = \{Y_i \mid i = 1, \dots, B\}$, are provided by pathologists or extracted from pathology reports. Typically, blank background areas in a WSI are masked out using Otsu's method [33] or other foreground segmentation techniques [34], and the remaining areas are divided into S_i patches, each referred to as an instance $\mathbf{x}_j^{raw}$. Accordingly, each WSI containing S_i patches is represented as a bag, $X_i^{raw} = \{\mathbf{x}_{ij}^{raw} \mid j = 1, \dots, S_i\}$.

(b) Multiple instance learning method. The extracted patches (with raw pixel values) are transformed into computer-processable representations via the feature extraction backbone $E(\cdot)$, where $E(X_i^{raw}) = X_i = \{\mathbf{x}_{ij} \mid j = 1, \dots, S_i\}$, $\mathbf{x}_{ij} \in \mathbb{R}^m$. Generally, converting a patch into a single score ($m = 1$) is known as the **instance-based MIL approach**, whereas converting a patch into a vector ($m > 1$) is known as the **embedding-based MIL approach** (or representation-based MIL approach) [35]. Regardless of the chosen MIL approach, the patch (instance) representations are aggregated to form a comprehensive representation of the entire WSI (bag). This involves defining an aggregation strategy that consolidates the set of patch representations $\{\mathbf{x}_{ij} \mid j = 1, \dots, S_i\}$ into a WSI-level representation $\mathbf{Z}_i \in \mathbb{R}^n$.

(c) Downstream tasks. The WSI representation $\mathbf{Z}_i$ effectively captures comprehensive information from billions of pixels and can be utilized for various downstream tasks such as classification, segmentation, and regression. For instance, in the classification task, a mapping function $F(\cdot)$ is defined to map $\mathbf{Z}_i$ to the label Y_i.

3. Trends of MIL in recent years

The annual number of publications on MIL has shown a significant increase since 2019, reaching an estimated total of 51 in 2024, which surpasses the cumulative number of publications from 2005 to 2018 (50) [29,36]. Notably, the publication growth rate peaked in 2021 with an unprecedented increase of 142%, marking the most substantial rise over the past six years. This section provides a detailed analysis of trends in MIL for CPATH from 2019 to the present (up to July 24, 2024), encompassing backbone architectures, pretraining datasets, optimization techniques, aggregation approaches, and downstream tasks. Insights are derived from a comprehensive review of 190 papers, with the findings illustrated in Fig. 2.

3.1. Backbone architectures

The trends in backbone architectures are illustrated in the top-left of Fig. 2. This analysis highlights the foundational methods employed by researchers for image feature extraction. The results indicate that ResNet has been utilized as a backbone in over 50% of studies conducted in the past six years. This dominance can be attributed to ResNet's pioneering approach in addressing network depth challenges, thereby enabling the extraction of deeper features—an essential advantage when processing billion-pixel WSIs in CPATH. Other CNN-based architectures, including DenseNet [37], EfficientNet [38], Inception [39], Xception [40], MobileNet [41], VGG [42], and Shufflenet [43], were widely adopted around 2019 due to their respective strengths, achieving a usage share comparable to ResNet. However, since 2021, the rapid advancement of Transformer technology has significantly altered this landscape. The share of other CNN-based methods has declined sharply, whereas Transformer-based backbones have gained increasing prominence annually. Currently, non-ResNet CNN-based methods account for only 11% of the total share, reflecting a dramatic 78% decrease over five years.

The choice of backbone plays a critical role in determining the effectiveness of feature extraction from images. Trends over the past six years reveal a clear evolution in researchers' preferences, with some favoring the well-established ResNet and others embracing the emerging Transformer methods [44,45]. While these two mainstream approaches dominate the field, other backbones are employed primarily to address specific challenges (e.g., Juyal et al. [46]'s work on imbalanced classification and OOD problems in MIL) or to tackle new clinical problems rather than algorithmic ones (e.g., Cai et al. [47]'s use of MIL for diagnosing and predicting the recurrence of odontogenic keratocyst). This shift underscores that the mainstream choice of backbone has transitioned from selecting among various CNN-based models to a dichotomy between ResNet and Transformer, marking a pivotal decision for researchers in this domain. Moreover, the trends in Transformer and ResNet adoption suggest that many methods in other areas may still require updates to align with these advancements.

3.2. Pretraining datasets

This section discusses three primary pretraining datasets: **ImageNet**, **histopathology**, and **no pretraining**. The results are displayed on the top-middle of Fig. 2.

Since its introduction in 2009, ImageNet has had a profound impact on deep learning, driving advancements across the entire computer vision domain. Consequently, most MIL methods adopted ImageNet for pretraining following its release. Although the domain gap between natural scene images and histopathological images can compromise the performance of pre-trained models on histopathological data [48], studies published in 2019 demonstrated that various backbones pre-trained on ImageNet were already sufficient for feature extraction from WSIs to accomplish numerous tasks [49–53]. In 2019, the three main pretraining datasets—ImageNet, Histopathology, and other approaches—were relatively evenly distributed, reflecting researchers' exploration

of diverse strategies. These included employing mature backbones pretrained on ImageNet, pretraining with large domain-specific datasets, or adopting alternative approaches such as end-to-end methods or leveraging data from other fields.

Following the validation of ImageNet-pretrained backbones for CPATH applications, their usage proportion increased further in 2020 and 2021, maintaining a consistently high share. However, from 2022 onwards, the proportion of studies utilizing Histopathology data for pretraining rose significantly, resulting in a noticeable decline in the reliance on ImageNet. This shift is largely attributed to the emergence and maturation of self-supervised and contrastive learning methods [25,26,54–57], which have been successfully applied in pathology [58,59], leveraging high-quality datasets such as TCGA [60] and Camelyon 16 [18].

Publications have consistently shown that contrastive learning methods, regardless of the backbone used, can effectively extract features. These methods are better suited to the CPATH domain than ImageNet and do not require additional annotations [61–64]. For instance, Xu et al. [65] demonstrated that backbones pretrained with Histopathology data outperformed those pretrained on ImageNet in feature extraction for pathological data, excelling in large-scale WSI datasets and multimodal tasks. While models pretrained on ImageNet can still facilitate rapid development for simpler downstream tasks, contrastive learning is the preferred choice for tasks involving data imbalance [66], insufficient representation capabilities of conventional methods [67,68], or specialized downstream applications [69]. These challenges are likely to persist in the future development of CPATH, suggesting that methods pretrained on both Histopathology data and ImageNet will continue to be mainstream.

The concentration of such studies in 2021 highlights the exploratory nature of feature extraction techniques. For example, Gao et al. [70] achieved subtyping of Papillary Renal Cell Carcinoma, and Lerousseau et al. [71] utilized the spatial relationships of tiles for whole-slide image classification. These studies reflect the ongoing exploration of methods that address the limitations of ImageNet-pretrained models in representation learning.

3.3. Optimization techniques

This section covers two model optimization methods: **freeze** and **unfreeze**. The results are displayed on the top-right of Fig. 2. Whether a model employs a freezing strategy for subsequent fine-tuning after pretraining not only reflects its adaptability to various downstream tasks but also indicates whether the model's representational capacity meets the requirements for feature aggregation and downstream tasks in the MIL framework. It is important to note that the optimization techniques discussed here do not encompass differences in optimizers, as current research suggests that these factors are not significantly related to the MIL framework itself.

The analysis indicates that, overall, models tend to be frozen more often than unfrozen, suggesting that researchers generally prefer to use pretrained backbones without fine-tuning. Early studies have shown that for common pathology problems, using pretrained backbones directly can extract sufficient features to support downstream tasks [49–53,72,73]. In 2019 and the recent years (2023 and 2024), the proportion of unfrozen models remained low, with significant variations occurring in 2020 and 2021, reaching 33% and 38%, respectively. As noted in Section 3.2, 2020 and 2021 marked the years with the lowest proportion of backbones pretrained with Histopathology data. During this exploratory phase, more researchers tended to fine-tune backbones pretrained on ImageNet (works that train from scratch are clearly categorized as unfreeze [70,71,74–76]). This approach proved intuitive and has led to valuable work, such as Gao et al. [77]'s introduction of the Minimal Point-Based (Min-Point) annotation method for training unfrozen models.

However, the proportion of research fine-tuning backbones began to decrease in 2022, stabilizing around 10% in 2023 and 2024. This change does not imply that fine-tuning backbones in MIL is meaningless; rather, it indicates that the representational capacity of current pretrained models is sufficient for most downstream tasks, especially with the widespread adoption of large-scale pretrained models in medical imaging. For example, Chen et al. [78] utilized a large-scale pretrained ViT model for feature extraction in MIL. Large-scale pretrained models, which leverage vast domain-specific training data, have attained adequate representational capabilities [65], often obviating the need for additional fine-tuning.

Therefore, the observed trends over the past six years do not simply represent a freeze–unfreeze–freeze cycle but rather a transition from pretraining on ImageNet or small-scale Histopathology data to utilizing large-scale pretrained models. This transition addresses most researchers' challenges with MIL feature extractors, enabling them to focus more on subsequent work related to aggregators and downstream tasks.

3.4. Aggregation approaches

The aggregation approaches in MIL consist primarily of two components: **MIL approaches** (the methods used to define the representation form of instances) and **pooling approaches**.

This section covers three types of MIL approaches (**embedding-based**, **instance-based**, and **others**, as shown in the bottom-middle of Fig. 2) and six pooling approaches (**linear attention**, **self-attention**, **average pooling**, **top-k pooling**, **max pooling**, and **others**, as shown in the bottom-left of Fig. 2). It is important to note that all methods utilizing attention mechanisms but not incorporating the q, k, v mechanism [79] are collectively referred to as linear attention in this paper.

3.4.1. MIL approaches

Over the past six years, embedding-based approaches have consistently dominated, which is a reasonable outcome. As image feature extraction technology has advanced, most researchers prefer to retain the richer low-dimensional features of instances rather than represent them with a simple score. Moreover, with the growing popularity of end-to-end models in the medical imaging field [80], MIL methods have also shown a trend towards end-to-end development [81,82]. This trend has led to embedding-based MIL approaches being predominant in publications over the past six years. As we mentioned earlier, unlike the original objective of MIL, CPATH places less emphasis on classifying individual instances. Compared to MIL applications in other domains, CPATH primarily focuses on obtaining accurate bag-level predictions. This task preference may explain the dominant trend of bag-level prediction observed in this section.

A particularly notable situation occurred in 2021: there were no instance-based works at all. During this critical exploratory period, the mainstream approach among researchers was to focus on how to better extract and aggregate image features, a paradigm that heavily relied on embedding-based MIL approaches. The core issue here is that the aggregation methods for instance scores were largely fixed around 2021, making principled improvements in this area more challenging compared to representation aggregation.

However, the abandonment of instance-based approaches did not persist beyond 2021; instead, numerous instance-based publications re-emerged. Using 2021 as a demarcation point, pre-2021 works often employed more conventional aggregation methods, such as the average pooling used by Gao et al. [77] for solving cancerous region detection and subtyping in WSIs. Post-2021, instance-based approaches have explored a new path in CPATH by integrating instance-based methods with various other mechanisms [62,83]. The key to this shift lies in the advancement of feature extraction technology, which has increased

the reliability of scores extracted from patches, thereby compensating for the previously weaker representational capacity of instance aggregation. Coupled with the inherent advantages of instance-based approaches, such as good interpretability and excellent adaptability, this has allowed them to regain their former prominence after being undervalued in 2021. Furthermore, other MIL approaches have consistently accounted for approximately 10% of the total between 2020 and 2023, typically involving the combination of instance-based and embedding-based approaches [61,84].

3.4.2. Pooling approaches

The trend also indicates that researchers' preferences for pooling approaches have evolved from predominantly traditional pooling methods (average pooling, top-k pooling, and max pooling) to primarily attention-based pooling approaches by 2024. This trend peaked in 2023, and although the proportion of other methods slightly increased in 2024, traditional pooling approaches continued to decline. The overall trend over the past six years reveals a shift: MIL methods in CPATH are increasingly addressing problems that do not adhere to the SMI assumption. For problems such as lymph node metastasis [85] and prognosis analysis [86,87], the relationship between WSI labels and individual patch labels clearly does not follow the SMI assumption, which renders traditional pooling approaches theoretically unsound.

Although the linear attention method was introduced by Ilse et al. [88] in 2018, its proportion was only slightly above 20% in 2019 and 2020, which did not reflect its potential to become a mainstream approach. However, both attention-based pooling methods saw a significant increase in 2021, a shift largely driven by the influential work of Dosovitskiy et al. [23] in 2020. Additionally, key works such as those by Lu et al. [89] using linear attention and Li et al. [61] using self-attention significantly influenced the field and guided research directions. Theoretical advancements in 2021, as discussed in Sections 3.2 and 3.3, led to breakthroughs in feature extraction capabilities, prompting researchers to explore more advanced pooling approaches. In this context, traditional pooling methods could no longer effectively aggregate features with large amounts of low-dimensional information, making the shift to attention-based pooling approaches inevitable. Consequently, post-2021, attention-based pooling methods became dominant, with their proportion fluctuating around 70

In contrast to the rise of attention-based pooling approaches, traditional pooling methods were almost entirely neglected in 2021, with only Li et al. [90]'s work using average pooling in the final aggregation step. This marked the decline of traditional pooling approaches in 2021. However, a notable rebound occurred in 2022, demonstrating that not all tasks require the representational power of attention-based pooling approaches. Some representative works have shown that theoretical innovations and specific clinical problems can still benefit from traditional pooling approaches rather than relying solely on attention [62,91].

Before 2021, the proportion of other methods was similar to traditional aggregation methods, primarily consisting of improvements and combinations of traditional techniques. For example, Xu et al. [53] used Max–Max and Max–Min standards to select instances, and Yao et al. [52] achieved feature aggregation by concatenating representations followed by a fully connected layer. In 2024, the proportion of other aggregation methods rose again. Unlike the simple aggregation methods from six years ago, current works are exploring new aggregation techniques. For instance, Cai et al. [47] employed the Patch Likelihood Histogram (PLH) pipeline for feature aggregation, and Gu and Dao [92] combined Transformer and Mamba, where the instance feature aggregation stage consists of L stacked MIL-SSM modules, each comprising a Bi-SSM and a 2D-CAB block with a residual connection [93].

3.5. Downstream tasks

This section presents an analysis of downstream tasks in MIL, encompassing classification, regression, and segmentation. In this context, regression tasks include those related to survival risk prediction, gene expression intensity prediction, and similar endeavors. The trend is displayed at the bottom-right of Fig. 2.

The analysis reveals that classification tasks have consistently dominated the field, particularly reaching a peak in 2021. During this pivotal year, much of the research focused on exploring fundamental technologies in MIL for CPATH, such as feature extraction, model representation capabilities, model training, and instance aggregation. Since classification was the primary problem MIL was initially designed to address, the majority of the work during this period treated classification as the downstream task. Significant advancements were made in MIL technologies tailored for classification in 2021, providing crucial guidance for subsequent research. For instance, Li et al. [61] developed the Dual Stream MIL framework, Shao et al. [94] introduced a mature approach for incorporating Transformers into MIL and provided proof of convergence, and Sharma et al. [81] developed a clustering-based end-to-end MIL method.

As with previous sections, the year 2021 marks a pivotal moment, after which two major changes are observed: (1) The proportion of segmentation tasks significantly declined, and (2) The proportion of regression tasks began to increase.

Segmentation's share did not show any signs of recovery post-2021, a phenomenon rooted in the fundamental principles of MIL. The inherent logic of MIL offers excellent interpretability, whether through instance scoring or the combination of embeddings with attention mechanisms, enabling the derivation of highly accurate instance representations. Consequently, when the patch size is sufficiently small, segmentation of WSIs is automatically achieved during the MIL process. This level of segmentation is sufficient to address most clinical issues, leading to a decrease in researchers viewing MIL combined with segmentation as a task requiring separate attention. This unique approach to segmentation in MIL was compellingly demonstrated in the work by Lu et al. [89] published in 2021. Since then, the number of publications focusing primarily on segmentation has remained low, a trend likely to continue in the foreseeable future. The emergence of foundational models in pathology, such as the multimodal, vision-language task-agnostic foundational model CONCH proposed by Lu et al. [95], trained on 1.17 million image-caption pairs, and the UNI model by Chen et al. [78], trained on over 100 million images, exhibit such strong feature representation capabilities that WSI-level segmentation in MIL is unlikely to be viewed as a primary task going forward.

However, this does not imply that segmentation will disappear from MIL in CPATH. Beyond WSI-level segmentation, some specific clinical applications still require pixel-level segmentation. Performing pixel-level segmentation directly on WSIs necessitates treating each pixel as an instance, implying that each WSI would be divided into billions of instances—a practically unfeasible and highly costly approach. Therefore, applying MIL for pixel-level segmentation in localized WSIs remains a valuable research area. An example of ongoing research in this domain is the work by Cai et al. [96], who proposed HRMIL, combining Transformer and regional MIL aggregators to achieve pixel-level segmentation with performance competitive with fully supervised methods.

Unlike segmentation, the proportion of regression tasks saw a notable resurgence after the downturn in 2021. Regression tasks primarily include prognostic risk prediction and molecular biomarker prediction [97]. The resurgence in regression is largely driven by an increase in prognostic prediction tasks, reflecting another evolving direction of MIL in CPATH: applying MIL to continuous value prediction tasks in medicine. Traditional MIL work predominantly focused on classification and segmentation tasks achieved through instance classification.

Table 1
Trends before and after 2021.

Trends	Before 2021	After 2021
Backbone architectures (Section 3.1)	ResNet and other CNNs	ResNet and transformer
Pretraining datasets (Section 3.2)	ImageNet	ImageNet and histopathology
Optimization techniques (Section 3.3)	Freeze and unfreeze	Freeze
MIL approaches (Section 3.4.1)	Decline in instance-based methods	Growth in instance-based approach
Pooling approaches (Section 3.4.2)	Traditional pooling	Attention pooling
Downstream tasks (Section 3.5)	Classification and segmentation	Classification and regression

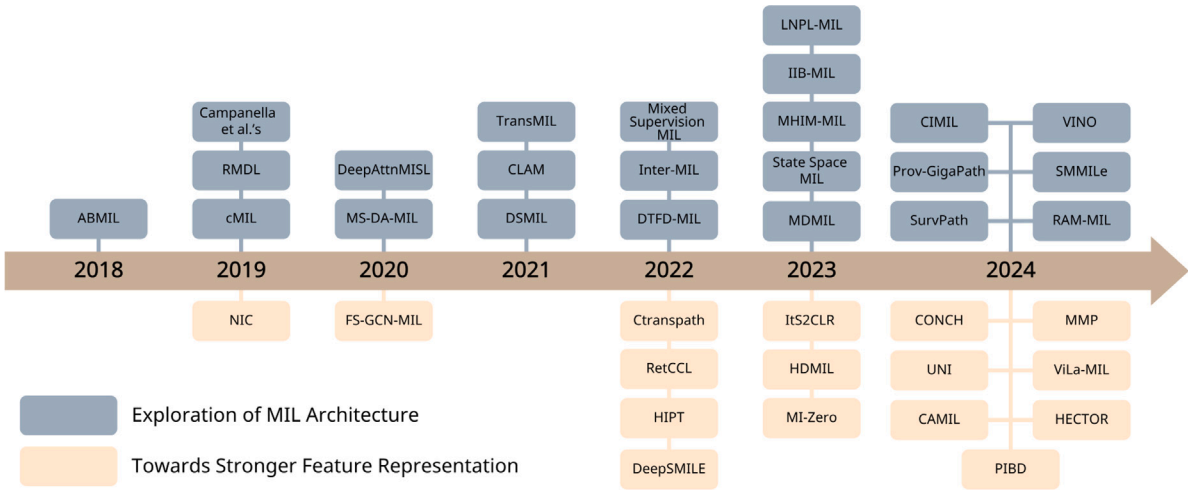

Fig. 3. Pivotal MIL works in CPATH from 2018 to 2024.

However, following the comprehensive exploration before and during 2021, there has been a gradual emergence of studies utilizing MIL for continuous value prediction.

A representative example of this trend is the work by Tsai et al. [98], who developed the Multi-omics Multi-cohort Assessment (MOMA) platform based on MIL. This platform successfully predicts overall survival, disease-free survival (log-rank test P-value < 0.05), and copy number alterations in CRC patients. Prognostic work is a typical multimodal task, where clinicians often need to integrate multidimensional information—such as electronic medical records, pathological slides, and pathology reports—to predict patient outcomes. This paradigm has been increasingly applied in MIL, especially in 2024, with the emergence of studies combining WSI with RNA sequencing data and pathology reports [99–101]. It can be anticipated that MIL methods in CPATH will increasingly be applied to prognostic prediction, particularly in conjunction with multimodal approaches.

Although molecular biomarker prediction is still relatively rare, the work by El Nahhas et al. [69] has already demonstrated the potential of MIL in this and related fields. As a result, MIL methods may see broader application in regression problems in the future, largely due to their excellent interpretability and ability to operate without relying on fine-grained annotations.

3.6. Summary

This section explores the development trends of MIL in CPATH from 2019 to 2024, with a critical turning point identified **around 2021**, as summarized in Table 1. This period marked significant shifts across various aspects of MIL, driven by two key factors: the emergence of groundbreaking works around 2021, which encompassed not only advancements in MIL architectures but also innovations in basic DL techniques; and the evolving demands of CPATH, which adapted alongside technological progress.

In particular, these transformative MIL architectures and DL techniques, building on foundational work from earlier years, set clear directions for the subsequent evolution of MIL techniques in CPATH. The rapid progress in MIL aggregation strategies, transformer-based backbones and self-supervised learning, particularly contrastive learning, after 2021 has profoundly shaped the observed trends. These advancements will be systematically reviewed in the next section.

4. Pivotal MIL studies in CPATH

This paper presents a review of the most representative MIL research in the CPATH field over the past six years, as illustrated in Fig. 3. These studies encompass significant methodological innovations, substantial clinical validations, or the introduction of new research directions. The selection criteria for these works include their publication date, the reputation of the journals or conferences, and the number of citations. Additionally, we have included works that, while not directly related to MIL techniques, have played a pivotal role in the development of the field.

These works are broadly categorized into two groups: **Exploration of MIL Architecture** and **Towards Stronger Feature Representation**. MIL methods, with their robust weakly supervised properties, are highly compatible with CPATH. As a result, improving their architecture to better leverage these capabilities has become a mainstream research direction. Given the nature of WSIs, it is unproductive to focus equally on a large number of irrelevant instances. Therefore, research in this area primarily focuses on selecting key instances and extracting the inherent relationships among them. Additionally, feature extraction techniques have remained a central focus of CPATH research. Building upon existing MIL frameworks, self-supervised learning (SSL) and multimodal fusion methods have been extensively explored and developed. The logical structure of this section is shown in Fig. 4.

4.1. Exploration of MIL architecture

Research on MIL architecture has consistently attracted significant attention, with a primary focus on optimizing the architecture to enhance various model performances. Landmark studies in this domain have laid theoretical foundations and provided empirical validation

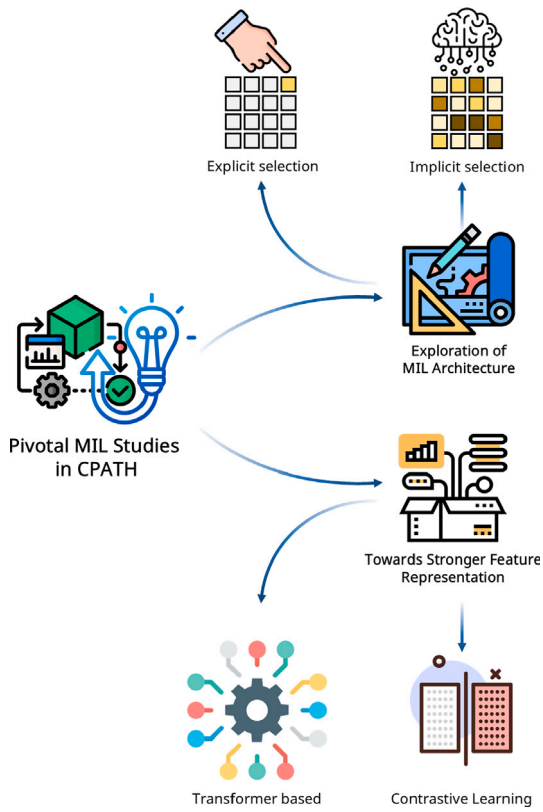

Fig. 4. The logical structure of Section 4.

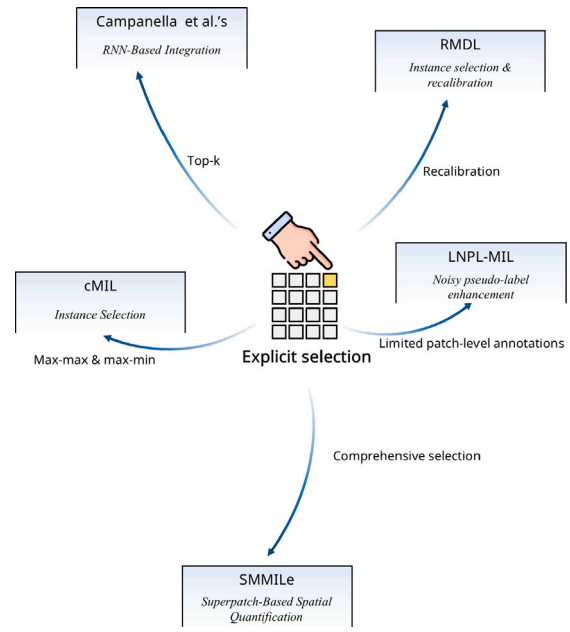

Fig. 5. Works involved in **Explicit selection**.

for subsequent research. These pioneering efforts have significantly expanded the influence of MIL methods in pathology. We categorize these works into two parts: **Valuable Instance Selection** and **Inter-instance Relationship Fusion**. The former emphasizes selecting more representative instances from a large pool, while the latter focuses on leveraging architectural advancements to explore relationships among instances. Further information of the works involved in this section is shown in Table 2.

4.1.1. Valuable instance selection

WSIs are often massive in scale, with a significant portion of their regions contributing little to the model. Given the limitations of computational resources, selecting the most valuable instances becomes crucial. Current approaches to valuable instance selection can be broadly categorized into two strategies: **explicit selection** and **implicit selection**. Explicit selection methods adopt a direct approach, identifying valuable instances through specific algorithms, while implicit selection methods delegate the determination of instance importance to the model itself.

In the early stages of explicit selection, straightforward algorithms were commonly employed. For example, Campanella et al. [49] employed a top-k instance selection strategy, feeding the selected instances into the subsequent aggregation stage to mitigate the computational burden caused by excessive token lengths. To further ensure the selection of the most valuable instances, **RMDL** [50] introduced a recalibration module to prevent misjudgment, and **cMIL** [53] proposed instance-based Max–Max and max–min selection methods. Over time, explicit selection techniques have been refined further. **LNPL-MIL** [103] leveraged limited patch-level annotations and pseudo-labels to assist in instance selection. Despite a decline in the popularity of explicit selection methods in recent years, **SMMILe** [104] demonstrated that instance-based explicit selection could still yield impressive results in spatial quantification tasks. Fig. 5 provides an overview of the works to be discussed in the following sections.

Campanella et al. [49] validated MIL on a dataset comprising 44,732 whole-slide images from 15,187 patients, achieving areas under the curve (AUC) greater than 0.98. Their approach enabled pathologists to exclude 65%–75% of slides while maintaining 100% sensitivity. This work strongly demonstrated the clinical applicability of MIL in CPATH. Three instance aggregation methods were employed: Naive multiscale aggregation, Random forest-based aggregation, and RNN-based aggregation. The latter also introduced learnable parameters, differing from ABMIL by selecting an RNN as the non-linear function and omitting the weighted average algorithm. However, it aligned with the underlying principles of ABMIL, providing experimental validation on large-scale clinical data.

RMDL [50] proposed Recalibrated Multi-instance Deep Learning to address the issue of oversized WSIs by adopting the conventional Region of Interest (ROI) paradigm, which extracts crucial patches from WSIs for diagnosis. To prevent the extracted patches from being unrepresentative, RMDL introduced a Recalibration module. This module aids the model in recalibrating the selected patches by fusing local and global features, ensuring that the patches chosen from a WSI are sufficiently representative.

cMIL [53] introduced the combined MIL and CAMEL model, marking a significant step in applying MIL to medical image segmentation. Although it did not achieve whole-slide image WSI-level segmentation, its approach to local image segmentation and the label enrichment method to address representational limitations have greatly influenced subsequent WSI-level and pixel-level segmentation techniques for localized medical images. Instead of adding extra labels for pixel-level segmentation, CAMEL divides WSIs into grids, generating labels for each local region during the enrichment process. This can be viewed as instance-level classification using WSI-level labels, with label enrichment enhancing dataset quality through combined MIL.

The cMIL method employs two instance selection criteria: Max–Max and Max–Min. Max–Max selects the instances with the highest positive scores for both positive and negative images. To balance the potential bias towards positive selections, Max–Min is introduced, offering a more balanced approach to instance selection. These criteria, when used independently, often produce opposite effects, addressing data imbalance issues.

Although not the first to explore this concept, the instance selection method in cMIL has become widely adopted in subsequent

Table 2
The works involved in Section 4.1. ('EMB' = 'embedding-based', 'INS' = 'instance-based', 'CLS' = 'classification', 'SEG' = 'segmentation', 'REG' = 'regression').

		Study	Year	Backbone	Pretraining dataset	Pooling approach	MIL approach	Downstream task	Brief highlights
Valuable instance selection	Explicit selection	Campanella et al.'s [49]	2019	ResNet	ImageNet	top-k	EMB	CLS	RNN-Based Integration
		RMDL [50]	2019	ResNet	ImageNet	Avg pooling	EMB	CLS	Instance selection & recalibration
		cMIL [53]	2019	ResNet	Histopathology	Others	INS	CLS, SEG	Instance selection
		Inter-MIL [102]	2022	ResNet	ImageNet	Linear attention	EMB	CLS, SEG	Self-interactive instance optimization
		LNPL-MIL [103]	2023	ResNet	ImageNet	top-k	EMB	CLS, REG	Noisy pseudo-label enhancement
		SMMILe [104]	2024	ResNet	ImageNet	Linear attention	INS	CLS	Superpatch-based spatial quantification
	Implicit selection	ABMIL [88]	2018	LeNet	No-pretraining	Linear attention	EMB	CLS	Attention pioneer in MIL
		DeepAttnMISL [105]	2020	CNN	ImageNet	Linear attention	EMB	CLS, REG	Applied attention to prognosis
		MS-DA-MIL [106]	2020	VGG	ImageNet	Linear attention	EMB	CLS	Multi-scale & domain adversarial
		CLAM [89]	2021	ResNet	ImageNet	Linear attention	EMB	CLS	Instance-level clustering
		DTFD-MIL [107]	2022	ResNet	ImageNet	Linear attention	EMB	CLS	Double-tier instance distillation
		MHIM-MIL [108]	2023	ResNet	ImageNet	Self attention	EMB	CLS	Masked hard instance mining
	–	DSMIL [61]	2021	ResNet	Histopathology	Self attention	Others	CLS	Dual-stream instance aggregation
Inter-instance relationship fusion	Transformer based	TransMIL [94]	2021	ResNet	ImageNet	Self attention	EMB	CLS	Introduced transformer to MIL
		Mixed supervision transformer [109]	2022	MobileNet	ImageNet	Self attention	Others	CLS	Mixed supervision fusion
		MDMIL [110]	2023	ResNet	ImageNet	top-k	Others	CLS	Multiplex detection for heterogeneity
		IIB-MIL [111]	2023	EfficientNet	ImageNet	Self attention	EMB	CLS	Integrated supervision with label disambiguation
		RAM-MIL [112]	2024	ResNet	ImageNet	Self attention	EMB	CLS	Cross-domain retrieval and interpretation
		VINO [113]	2024	ResNet	ImageNet	Avg pooling	EMB	CLS	Transformer-based video-structure
		Prov-GigaPath [65]	2024	ViT	Histopathology	Linear attention	EMB	CLS	Relationship fusion with dilated attention
	–	State space MIL [114]	2023	ResNet	ImageNet	Others	Others	CLS	State space models for Long-sequence
	–	CIMIL [115]	2024	ResNet	ImageNet	Others	INS	CLS, SEG	Counterfactual boosting

MIL research [77,116]. While it only achieved ROI-level segmentation due to limitations in feature extraction and computational power, its contribution was pivotal in advancing the field.

LNPL-MIL [103] introduced Learning from Noisy Pseudo Labels to enhance MIL, a method similar to Mixed Supervision MIL [109]. Both approaches focus on optimizing feature extraction in scenarios with limited patch-level annotations (LPA) [103]. While using LPA to assign pseudo labels to unlabeled patches offers a practical guide for MIL, it often introduces noise. Furthermore, training MIL on WSI-level labels can suffer from semantic misalignment between instances and bag-level labels. To address these issues, LNPL-MIL introduces the Super-Patch-based LNPL (SP-LNPL) and Transformer aware of instance Order and Distribution (TOD-MIL). LNPL-MIL uses LPA to improve feature extraction while reducing the negative impact of noise. Although this approach somewhat contradicts the core principles of MIL, it suggests a promising direction for improving MIL's feature extraction by expanding network scale and incorporating small amounts of additional information to enhance model performance.

SMMILe [104] proposed the Superpatch-based Measurable MIL method, a key contribution in instance-based MIL studies post-2021. Spatial quantification is critical in CPATH tasks, such as guiding pathologists and identifying tissue phenotypes linked to novel biomarkers. A major challenge in developing spatially-aware CPATH models is the need for detailed spatial annotations, which are difficult to obtain due to the large scale of gigapixel images and the requirement for specialized domain knowledge [117]. Embedding-based MIL methods often produce imprecise attention maps, limiting their effectiveness in spatial analysis [118,119]. This constraint hampers their use in tasks requiring accurate phenotypic descriptions rather than global labels. Gao et al. [104] demonstrated mathematically that instance-level aggregation can achieve superior spatial quantification without sacrificing WSI prediction performance. SMMILe integrated the NIC [72], two novel instance-based attention modules, and a Bayesian refinement network. The significance of this work lies in its rigorous mathematical comparison of instance-based and embedding-based MIL methods, highlighting the advantages of the former in spatial quantification. This could pave the way for further research in this area.

Implicit selection methods for MIL in the CPATH field originated with **ABMIL** [88], which introduced attention mechanisms to allow the model to autonomously handle instance selection. This approach inspired a series of related advancements. **DeepAttnMISL** [105] applied this method to prognosis tasks, while **MS-DA-MIL** [106] extended ABMIL by incorporating multi-scale feature extraction and domain-adversarial techniques for improved selection. **CLAM** [89] added instance-level clustering, **Inter-MIL** [102] built upon attention mechanisms and encoder fine-tuning, not only selecting instances with higher attention scores but also considering supplementary and non-attentive instances, **DTFD-MIL** [107] introduced a double-tier MIL framework for instance feature distillation, and **MHIM-MIL** [108] implemented a Siamese structure to facilitate feature selection through a teacher–student architecture. Fig. 6 provides an overview of the works to be discussed in the following sections.

ABMIL [88] introduced Attention-based MIL pooling in 2018, presenting a novel instance aggregation pooling method, referred to as linear attention in Section 3 of this paper. Prior to this work, MIL methods commonly employed various operators for instance aggregation, such as the maximum operator, mean operator, and convex maximum operator [120]. These differentiable operators, favored for their computational simplicity and interpretability, were routinely used for instance aggregation in MIL. However, a significant limitation of

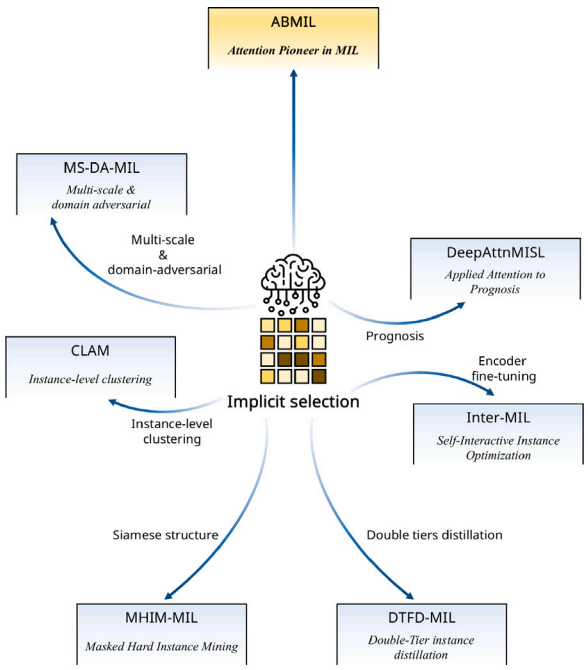

Fig. 6. Works involved in **Implicit selection**.

these traditional pooling methods is that they are predefined and cannot be optimized during subsequent training. This drawback became particularly pronounced with the rapid advancements in deep learning in 2018.

Moreover, the adaptability of these traditional pooling methods to embedding-based approaches is problematic. Aggregating instance embeddings using the max-operator is evidently unreasonable, while employing the mean operator, although yielding a bag representation, often results in suboptimal performance. As discussed in Section 3.4, the dominance of embedding-based MIL approaches within the field necessitated the development of a more flexible, trainable, and adaptive instance aggregation method, ideally one that also retains interpretability.

ABMIL introduces a weighted averaging method where the weights are learned through neural network training. Given a bag $X = \{\mathbf{x}_1, \dots, \mathbf{x}_S\}$ containing S embeddings, ABMIL is represented as follows:

$$\mathbf{z} = \sum_{k=1}^{S} a_k \mathbf{x}_k \tag{1}$$

where:

$$a_k = \frac{\exp\{\mathbf{w}^\top \tanh(\mathbf{V}\mathbf{x}_k^\top)\}}{\sum_{j=1}^{S} \exp\{\mathbf{w}^\top \tanh(\mathbf{V}\mathbf{x}_j^\top)\}} \tag{2}$$

where $\mathbf{w} \in \mathbb{R}^{l\times 1}$ and $\mathbf{V} \in \mathbb{R}^{l\times n}$ are parameters. Due to the fact that the tanh non-linearity is nearly linear over the interval $[-1, 1]$, it can be inefficient for learning complex relationships. To address this issue, a second non-linearity can be selectively introduced. This approach employs a gating mechanism [121], which is used in conjunction with the tanh non-linearity, resulting in the following formulation:

$$a_k = \frac{\exp\{\mathbf{w}^\top[\tanh(\mathbf{V}\mathbf{x}_k^\top) \odot \mathrm{sigm}(\mathbf{U}\mathbf{x}_k^\top)]\}}{\sum_{j=1}^{S} \exp\{\mathbf{w}^\top[\tanh(\mathbf{V}\mathbf{x}_j^\top) \odot \mathrm{sigm}(\mathbf{U}\mathbf{x}_j^\top)]\}} \tag{3}$$

where $\mathbf{U} \in \mathbb{R}^{l\times n}$ are parameters, $\odot$ denotes element-wise multiplication, and $\mathrm{sigm}(\cdot)$ is the sigmoid non-linearity.

The gating mechanism effectively addresses the linearity issue associated with the tanh function. Notably, this weighted average pooling method can be interpreted as a form of attention mechanism [122,123]. Although termed "linear attention", the calculation of instance attention scores actually employs two non-linearities. This approach utilizes a trainable neural network to capture more complex interactions within the image. The essence of this attention mechanism lies in its ability to transform instance feature vectors into attention scores. The simplicity of this combination allows the introduction of trainable parameters while preserving the interpretability of traditional pooling methods.

DeepAttnMISL [105] introduced Deep Attention Multiple Instance Survival Learning, which was the first work to combine attention mechanisms with MIL for patient survival prediction. Unlike other studies, DeepAttnMISL treated a patient as a bag, which could contain multiple WSIs. The authors employed clustering to collect instances and used Siamese MI-FCN to learn features from phenotype clusters (similar to Yao et al. [52]'s previous work). In the feature aggregation stage, DeepAttnMISL used a MIL pooling method consistent with ABMIL.

MS-DA-MIL [106] introduces the Multi-scale Domain-Adversarial MIL, which integrates MIL with domain adversarial and multi-scale learning techniques. To address variations in staining conditions, MS-DA-MIL uses a domain-adversarial neural network $f_d(\cdot \mid \theta_d)$, where θ_d represents the trainable parameters. This method effectively disregards irrelevant domain differences that do not contribute to the classification task, outperforming traditional techniques like color augmentation and stain normalization [124].

In the context of histopathological image classification, MS-DA-MIL treats each patient as a separate domain, thus ignoring staining differences between patients. Initially, the feature extractor is trained at a single scale using DA to ignore staining variations in WSIs. Subsequently, the multi-scale DA-MIL network is trained, aggregating instance embeddings from different scales into a bag-level representation using attention scores from ABMIL.

MS-DA-MIL is the first framework to combine multi-scale, domain-adversarial, and MIL methods. It offers accurate cancer subtype classification while addressing staining variability, marking a significant contribution to the field. Despite challenges in selecting instances across scales, its pioneering approach to simulating the diagnostic process of pathologists has influenced subsequent research [125–128].

CLAM [89] introduces the Clustering-constrained-attention MIL, addressing poor domain adaptation and interpretability in CPATH by leveraging clustering during training. CLAM uses attention-based learning to identify subregions with high diagnostic value and applies instance-level clustering to refine the feature space.

The feature extraction and aggregation process in CLAM is similar to that of Ilse et al. [88], with the key modification being the introduction of a class-specific weight $\mathbf{w}$ in Eq. (3), generating distinct bag representations for each class. CLAM further incorporates a binary clustering objective to enhance class-specific feature learning. For each class, a fully connected layer $\mathbf{W}_{inst}$ predicts cluster assignment scores $\mathbf{p}_k = \mathbf{W}_{inst,k}\mathbf{x}_k$ for the kth patch. During training, the attention branches are supervised by the WSI-level label. Instances with high attention scores provide strong positive evidence for class Y, while low scores indicate strong negative evidence. If classes are mutually exclusive [129], the attention network's output, trained with WSI labels, can be used as pseudo labels for clustering.

CLAM has been successfully applied to detecting renal cell carcinoma, non-small-cell lung cancer, and lymph node metastasis. Experimental results confirm CLAM's reliability and generalizability, demonstrating the effectiveness of MIL methods in processing WSIs with low cost and improved interpretability. This approach also provides a foundation for future works in the field [81,130–132].

Inter-MIL [102] proposed the self-interactive MIL, which iteratively exchanges training information between fine-grained and global context features. The process is divided into three steps: (1) The AttPool network computes instance embeddings and obtains attention scores for

each instance; (2) An instance-level feature pool is constructed using high-attention instances, supplementary instances, and non-attention instances; (3) The instance-level feature pool is used for fine-grained training of the CNN backbone encoder, generating a new batch of instance embeddings for AttPool training in the subsequent epoch.

This method essentially groups instances based on their attention distribution, utilizing this information during the instance aggregation stage to provide feedback into the training of the feature extractor. This feedback improves the separation between positive and negative instances, ultimately enhancing the overall model performance. However, Inter-MIL has not gained widespread adoption in subsequent research, likely due to its complex architecture, limitations on the number of patches, and less-than-ideal interpretability.

DTFD-MIL [107] proposed the Double-Tier Feature Distillation MIL approach, which addresses the issue of processing a large number of patches, a problem primarily caused by the imbalance between positive and negative instances. The DTFD-MIL framework adopts a hierarchical instance filtering strategy. It introduces the concept of a pseudo-bag, where a parent-bag is divided into multiple pseudo-bags. Each pseudo-bag inherits the parent-bag's label and contains a subset of patches. The first step involves applying Attention-Based MIL (ABMIL) to all pseudo-bags. Then, a second-tier ABMIL model is employed to address label errors in pseudo-bags that do not contain positive instances, based on the features extracted from all pseudo-bags.

Unlike approaches that heavily rely on mutual-instance relations [61,94,133], DTFD-MIL partially mitigates the imbalance between positive and negative instances in WSIs by increasing the model's scale. This method offers a novel perspective and demonstrates strong performance in addressing these challenges.

MHIM-MIL [108] introduces the Masked Hard Instance Mining MIL, addressing the issue of unbalanced positive and negative instances, which causes traditional MIL models to focus on easily classified instances and neglect harder ones. MHIM-MIL combines a Siamese structure with instance masking strategies to implicitly mine hard instances through attention, improving model discriminability. Inspired by Li et al. [134], it masks simple instances with high attention scores to emphasize harder ones.

MHIM-MIL integrates High Attention Masking (HAM) with two additional strategies—lowest and random masking—to form LR-HAM. It uses a momentum teacher in the Siamese structure to compute attention scores for all instances, helping the student model focus on harder instances. This framework can be applied to existing attention-based MIL models, such as ABMIL, DSMIL, and TransMIL, enabling more efficient training and faster convergence by focusing on informative regions and mitigating underfitting.

Additionally, **DSMIL** [61] combined instance-based and embedding-based approaches for feature selection, representing a hybrid method that bridges explicit and implicit strategies.

DSMIL [61] introduced the Dual-stream MIL to address two key challenges in MIL: (1) class imbalance, which can lead to overfitting by neglecting positive instances [88], and (2) the high cost of end-to-end training, which can reduce the feature extractor's representational power [49,51]. DSMIL overcomes these challenges by combining max-pooling and self-attention in its instance aggregator. Key innovations include using SimCLR [25], a self-supervised contrastive learning method, for training the feature extractor, and employing a dual-stream instance aggregator.

Unlike the multiscale method in MS-DA-MIL [106], DSMIL combines features from different scales before aggregation. This Locally-Constrained Multiscale Attention mechanism has two advantages: (1) low-magnification representations share similar attention weights, and (2) it retains information from multiple scales, allowing the network to focus on relevant features across them. Although similar to self-attention [79], it differs by performing query-key matching only between the critical instance and others, including itself. This quasi-self-attention mechanism assigns higher attention to instances closer to the critical one, making it effective in pathology tasks where classification often relies on small local regions, while avoiding the computational cost of strict self-attention.

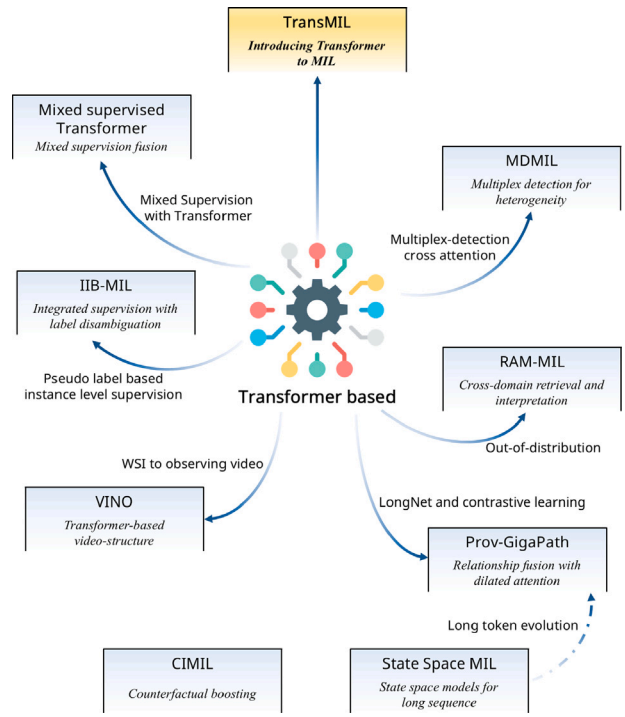

Fig. 7. Works involved in **Inter-instance Relationship Fusion**.

4.1.2. *Inter-instance relationship fusion*

Beyond extracting valuable instances, mining the rich contextual information within WSIs is another critical aspect of research. In the context of MIL, this effort manifests as the extraction of inter-instance relationships, referred to as inter-instance relationship fusion. Due to their powerful ability to efficiently capture contextual relationships, Transformers have been widely adopted in this research area. **TransMIL** [94] was the first to introduce Transformers into MIL. While its overall workflow followed the implicit feature selection paradigm, TransMIL diverged from the MIL pooling approach used in ABMIL by focusing on extracting inter-instance relationships rather than selecting valuable instances.

Building on TransMIL, subsequent works have made significant advancements. For instance, Bian et al. proposed **Mixed Supervision Transformer** [109] to enhance model performance. **MDMIL** [110] simplified traditional multi-head self-attention by employing multiplex-detection cross-attention, alleviating the computational burden of constructing contextual connections. **IIB-MIL** [111] introduced instance-level supervision using pseudo-labels. **RAM-MIL** [112] introduced the Optimal Transport (OT) distance and was the first systematic exploration of the out-of-distribution (OOD) problem in MIL. **VINO** [113] extended the Transformer-MIL paradigm to pathological video analysis, broadening its application scope. **Prov-GigaPath** [65] enabled Transformer-based MIL to handle ultra-long token sequences by incorporating LongNet, pushing the boundaries of what these models can achieve in processing extended sequence data. Fig. 7 provides an overview of the works discussed below.

TransMIL [94] proposed the Transformer-based Correlated MIL, which addresses the limitations of traditional methods that rely on the i.i.d. assumption, often unrealistic in clinical settings [135]. Pathologists typically consider both specific regions and their surrounding context, making inter-instance correlations crucial for accurate diagnosis.

TransMIL leverages the Transformer's self-attention mechanism, which models pairwise relationships between all tokens in a sequence, in contrast to the weighted average approach used in ABMIL. While

traditional Transformers are computationally intensive, limiting their application to large-scale images like WSIs, TransMIL overcomes this challenge with the TPT module, which includes the following steps: (1) sequence squaring, (2) correlation modeling, (3) conditional position encoding and local fusion, (4) deep feature aggregation, and (5) WSI-level prediction. The TPT module operates on instance embeddings extracted using ResNet-50, transforming feature bags into predicted labels. To mitigate the high computational cost of self-attention, TransMIL uses the Nyström Method [136] to approximate self-attention in WSIs.

TransMIL effectively integrates self-attention into MIL, addressing the mismatch between the i.i.d. assumption and real-world diagnostic workflows. While it does not fully resolve the issue of extremely long sequences in WSIs, it provides a viable approach for future MIL methods utilizing self-attention.

Mixed Supervision MIL [109] represents a typical application of mixed supervision in MIL. In conventional MIL methods, considering only WSI-level labels may cause the model to overlook rich local information within the image. Conversely, incorporating limited pixel-level labels may negatively impact model performance due to the potential inaccuracy of these labels. To address this issue, mixed supervision MIL proposed three main innovations: (1) Superpixel-based Instance Feature and Label Generation: This technique converts limited pixel-level labels into instance-level labels, enabling the model to leverage more granular information; (2) Random Masking Strategy: This strategy is used to prevent inaccurate instance-level labels from negatively affecting the model's performance, mitigating the risks associated with label noise; (3) 2D Sinusoidal Spatial Position Encoding: This method encodes the spatial position information of instances, helping the model incorporate spatial context more effectively.

Although mixed supervision MIL utilizes limited pixel-level labels from the dataset, its approach of expanding the network to generate and calibrate instance-level labels has been adopted in subsequent research. This innovation bridges the gap between WSI-level and instance-level supervision, improving overall model performance by integrating finer-grained information into the MIL framework.

MDMIL [110] introduced Multiplex-detection-based MIL to address the challenges posed by tumor heterogeneity on MIL performance. The key contributions of MDMIL are as follows: (1) Internal Query Generation Module (IQGM): The IQGM aggregates highly reliable instances through a classifier layer and probability distribution analysis to generate an internal query; (2) Multiplex Detection Module (MDM): Comprising multiplex-detection cross-attention and multi-head self-attention, the MDM generates bag-level representations. The multiplex-detection strategy utilizes both internal and variational queries, which function complementarily, significantly enhancing the model's instance mining and generalization capabilities; (3) Memory-Based Contrastive Loss: Inspired by MoCo [137], memory-based contrastive loss ensures consistency across various phenotypes in the feature space, mitigating the challenges posed by tumor heterogeneity and enhancing training stability. MDMIL's modular innovations effectively address the performance degradation caused by tumor heterogeneity. However, its primary drawback lies in the potential loss of contextual information during computation. This limitation renders MDMIL less applicable to most real-world clinical pathology tasks, where preserving contextual details is crucial.

IIB-MIL [111] introduced Integrated Instance-Level and Bag-Level MIL, which combines bag-level and instance-level supervision to address the issue of the lack of instance-level labels. The key innovation of IIB-MIL is the Label Disambiguation Module, designed to correct imprecise labels assigned to instances, enabling instance-level supervision. This module is essential for enhancing the supervision available at the instance level. In IIB-MIL, extracted instance embeddings are passed through a Residual Transformer Backbone. The Label Disambiguation Module then generates pseudo-labels, which are used to calculate the loss in conjunction with the output of the Instance Classifier, enabling the model to be optimized cooperatively. The Label Disambiguation Module consists of four main steps: (1) Obtain the instance classifier output; (2) Obtain the prototype labels; (3) Obtain soft labels from the Confidence Bank; and (4) Calculate instance-level loss. The core of IIB-MIL is overcoming MIL's inherent lack of supervision by obtaining additional supervision, significantly increasing the amount of training information available. However, this approach also adds complexity to the training process.

RAM-MIL [112] introduced Retrieval-Augmented MIL, inspired by Verma et al. [138], to address the performance degradation of MIL models when tested on an out-of-distribution (OOD) test set. The main contributions of RAM-MIL are: (1) employing optimal transport (OT) as a distance metric between bags [139]; and (2) demonstrating that reducing input dimensions leads to improved MIL performance. Although the proposed OT distance for MIL is limited by the number of instances within a bag, RAM-MIL, as one of the earliest works to study OOD performance in MIL, introduces a novel distance metric between bags, similar to the approach taken by FDD [140]. Additionally, it presents a RAM-MIL framework designed to address the OOD problem.

VINO [113] proposed a unique MIL framework to achieve end-to-end learning in MIL and capture contextual information. It divides WSIs into multiple patches, simulating the process by which pathologists review WSIs. These patches are modeled as a video reflecting the observation process. The video is then fed into a multi-head self-attention mechanism, directly obtaining both the clip token, which represents the semantic information of different categories, and the interactively computed clip feature. This novel approach mimics the diagnostic process of pathologists, moving away from conventional two-stage MIL methods. It enables end-to-end learning while enhancing the model's ability to capture contextual information.

Prov-GigaPath [65] developed a whole-slide pathology foundation model pretrained on 1.3 billion 256 × 256 pathology image patches from 171,189 WSIs obtained from Providence, a large U.S. health network comprising 28 cancer centers. This massive dataset includes over 30,000 patients and covers 31 major tissue types. GigaPath employs a vision transformer architecture for pretraining gigapixel pathology slides, with a significant innovation being the application of the LongNet method [141] to digital pathology. LongNet is a variant of the Transformer that achieves linear computational complexity and logarithmic dependency, allowing the extension of sequence token lengths to over one billion without compromising short-sequence performance. This architecture, well-suited for ultra-long sequences, aligns perfectly with the demands of CPATH. Prov-GigaPath follows the traditional MIL approach with feature extraction and feature aggregation architecture. The local feature extractor is pretrained using DINO v2 [142], the state-of-the-art image self-supervised learning (SSL) framework, while feature aggregation is achieved through masked autoencoder pretraining [143] and LongNet. By combining MIL with multiple advanced technologies and validating its performance on a vast dataset, Prov-GigaPath demonstrates outstanding results in various clinical downstream tasks, such as cancer subtyping and mutation prediction. It also demonstrates that MIL utilizing large-scale pretrained models is feasible for real-world clinical data in CPATH.

Although most outstanding works in this area are based on Transformers and their variants, some studies have explored alternative approaches. **CIMIL** [115] leveraged counterfactual reasoning and hierarchical instance searching to provide accurate pseudo labels for instances, thereby uncovering deeper inter-instance relationships. **State Space MIL** [114], on the other hand, addressed a similar problem to Prov-GigaPath, focusing on the application of MIL to long token sequences but employing structured state space models instead of Transformer architectures.

CIMIL [115] proposed a Counterfactual Inference-based framework to enhance MIL models, a model-agnostic framework designed to improve MIL at both the bag-level and instance-level. It introduces three key strategies: (1) counterfactual inference-based sub-bag assessment,

(2) hierarchical instance searching, and (3) feature refinement. In detail, the sub-bag assessment evaluates the impact of each sub-bag by removing it and observing the change in classification results, thereby improving the model's interpretability and classification performance. Hierarchical instance searching performs searches at both the sub-bag and instance levels, using the results from the previous step to accurately identify critical instances and generate precise pseudo-labels for subsequent instance-level classifier training. Feature refinement trains an instance-level classifier to generate instance embeddings, which are then used to replace the initial features, further enhancing the model's performance.

State Space MIL [114] marked the first use of Structured State Space Models (SSM) [144–146], an emerging sequence modeling method, as a MIL learner to efficiently model long sequences. SSMs are particularly effective for long-sequence modeling. Their experiments demonstrated that the combination of SSMs with MIL exhibits significant competitiveness across various downstream tasks.

4.2. Towards stronger feature representation

Stronger feature extraction techniques have consistently been a key pursuit in both CPATH and the broader computer vision field. MIL-based CPATH methods, by fully leveraging the advantages of MIL, enable feature extraction capabilities that surpass other methods, while also exhibiting high compatibility with various advanced feature extraction techniques. We categorize these works into two parts: **Self-Supervised Discoveries** and **Multimodal Insights Fusion**. The former combines MIL with SSL methods, achieving superior feature representations without the need for labels, thus enabling models to better address the data labeling challenges faced in the CPATH domain. The latter expands the data acquisition scope beyond WSIs, instead utilizing various available sources of information within the CPATH field, such as pathology reports and genetic sequencing data, to perform feature extraction through a multimodal approach. This enables the model to achieve feature extraction capabilities that exceed those of unimodal methods. Further information of the works involved in this section is shown in Table 3.

4.2.1. Self-supervised discoveries

The ability to leverage vast amounts of unlabeled data for pretraining is an attractive prospect for the CPATH field. Fortunately, MIL methods are highly compatible with self-supervised learning (SSL). The evolution of SSL in MIL closely follows temporal advancements. Prior to 2020, methods such as **NIC** [72] and **FS-GCN-MIL** [147] explored contrastive learning, variational autoencoders, and generative adversarial networks. Following the release of SimCLR [25] in 2020, the CPATH field began adopting contrastive learning frameworks, which had already gained recognition in deep learning and computer vision.

RetCCL [59], based on MoCo v2, introduced a distance-based clustering approach. Building on RetCCL, **CAMIL** [69] achieved regression tasks for continuous biomarkers directly from WSIs. **CTransPath** [58], extending MoCo v3, integrated Transformers to address challenges in assembling diverse features in SSL MIL frameworks. **HIPT** [148], utilizing DINO v1, introduced Hierarchical ViT, while **UNI** [78], based on DINO v2, pretrained models on over 100 million images.

Some studies have also deviated from directly applying mature frameworks to pathology-specific challenges [152]. A notable example is **ItS2CLR** [67], which aimed to enhance traditional contrastive learning methods in addressing class imbalance. This was achieved by leveraging instance-level pseudo labels derived from bag-level annotations, which enhanced the learned representations and captured inter-class differences.

In summary, the progression of SSL methods within MIL frameworks closely mirrors advancements in contrastive learning in deep learning and computer vision. As contrastive learning continues to evolve, it is anticipated that further groundbreaking works will emerge, applying

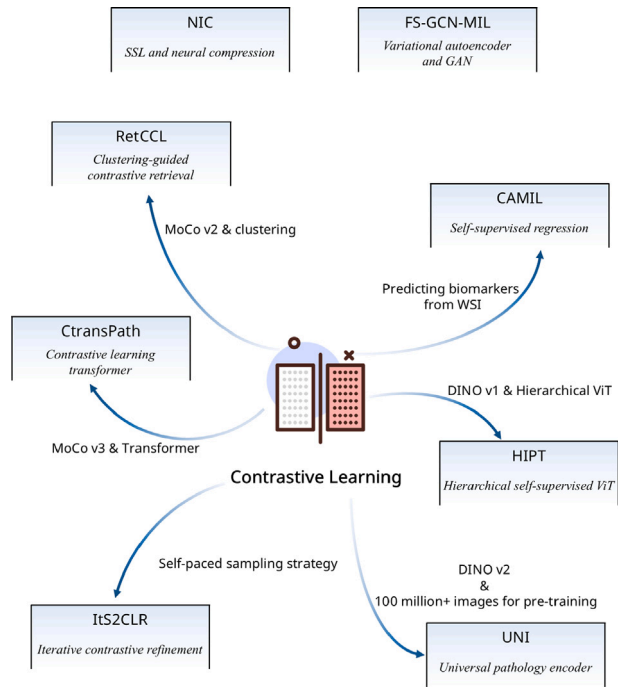

Fig. 8. Works involved in **Self-Supervised Discoveries**.

established frameworks to MIL and CPATH challenges. Fig. 8 provides an overview of the works to be discussed.

NIC [72] proposed Neural Image Compression to tackle the problem of oversized WSIs, drawing inspiration from cognitive mechanisms. Human observers can describe complex visual patterns with a few words rather than detailing individual pixels. This concept can be translated into deep learning, where the encoder uses low-dimensional embedding vectors to describe patches, ignoring irrelevant details. Compared to patch selection methods, compressing WSIs theoretically ensures the preservation of critical information through training, achieving significant compression rates [153]. NIC implemented WSI compression using three methods: Variational Autoencoder, Contrastive Training, and Bidirectional Generative Adversarial Networks, representing a cutting-edge attempt in 2019. Unfortunately, NIC did not combine these methods, and at that time, deep learning feature extraction capabilities were not advanced enough to support an end-to-end approach like NIC for obtaining bag representations directly from WSIs.

FS-GCN-MIL [147] proposed Feature-selection GCN MIL, which introduced a method that integrates Variational Autoencoder (VAE) [154], Generative Adversarial Network (GAN) [155], feature selection (FS), and Graph Convolutional Network (GCN) into MIL, addressing challenges in large-scale WSIs and the high cost of annotations for lymph node metastasis prediction.

The FS-GCN-MIL pipeline includes image preprocessing, instance-level feature extraction, feature selection, and bag-level classification. VAE-GAN is used for feature extraction, followed by FS to identify discriminative features. GCN aggregates instance embeddings for binary prediction. This method requires pathologists to manually annotate regions of interest (ROIs) for tumor patches. To address the challenge of insufficient training data in medical imaging, FS-GCN-MIL uses feature selection to reduce redundancy by selecting the most discriminative instance-level features, simplifying the model's learning task. Histogram-based techniques bridge instance-level and bag-level features, with maximum mean discrepancy [156] used to evaluate instance importance.

FS-GCN-MIL was one of the first to apply self-supervised learning (SSL) for feature extraction in MIL. Its combination of feature selection

Table 3
The works involved in Section 4.2. ('EMB' = 'embedding-based', 'INS' = 'instance-based', 'CLS' = 'classification', 'SEG' = 'segmentation', 'REG' = 'regression').

		Study	Year	Backbone	Pretraining dataset	Pooling approach	MIL approach	Downstream task	Brief highlights
Self-supervised discoveries	–	NIC [72]	2019	CNN	Histopathology	Avg pooling	EMB	CLS	SSL and Neural compression
	–	FS-GCN-MIL [147]	2020	ResNet	Histopathology	Others	EMB	CLS	Variational autoencoder and GAN
	Contrastive learning	Ctranspath [58]	2022	Swin transformer	Histopathology	–	–	Task agnostic	Contrastive learning transformer
		RetCCL [59]	2022	ResNet	Histopathology	–	–	Task agnostic	Clustering-guided contrastive retrieval
		HIPT [148]	2022	ViT	Histopathology	–	–	Task agnostic	Hierarchical self-supervised ViT
		ItS2CLR [67]	2023	–	Histopathology	–	–	Task agnostic	Iterative contrastive refinement
		UNI [78]	2024	ViT	Histopathology	–	–	Task agnostic	Universal pathology encoder
		CAMIL [69]	2024	ResNet	Histopathology	Linear attention	EMB	REG	Self-supervised regression
Multimodal insights fusion	Modality alignment	PLIP [149]	2023	ViT	Histopathology	–	–	Task agnostic	Twitter-powered medical insights
		MI-Zero [64]	2023	ViT	Histopathology	–	–	CLS	Large-scale histopathology understanding
		CONCH [95]	2024	ViT	Histopathology	–	–	Task agnostic	Pretrained vision-language for pathology
		ViLa-MIL [150]	2024	ResNet	Histopathology	Self attention	EMB	CLS	Dual-scale vision-language model
	Modality aggregation	MMP [101]	2024	ViT	Histopathology	Self attention	EMB	REG	Multimodal fusion for cancer prognosis
		HECTOR [151]	2024	ViT	Histopathology	Linear attention	EMB	CLS	Integrating pathology, RNA, and Clinical data
		PIBD [100]	2024	Swin Transformer	Histopathology	Self attention	EMB	REG	Addressing intra- and inter-modal redundancy
		SurvPath [99]	2024	ViT	Histopathology	Self attention	EMB	REG	Dense multimodal interaction model

and GCN significantly advanced MIL in cancer lymph node metastasis [84,85,157–160], demonstrating that MIL can perform well even when the SMI assumption is not met. Although the use of ImageNet pretraining peaked in 2020, FS-GCN-MIL highlighted the potential of models specifically trained on histopathology data. Subsequent research has reduced the need for selective feature vector pruning, but similar techniques for instance importance evaluation remain prevalent [47,128]. Moreover, FS-GCN-MIL's integration of GCN and self-attention has had a significant impact on studies of spatial feature extraction in pathological images [161–164].

RetCCL [59] introduced the Retrieval with Clustering-guided Contrastive Learning framework for robust and accurate WSI-level image retrieval. RetCCL consists of two stages: Stage A, a CCL-based feature extractor, and Stage B, a ranking and aggregation algorithm for WSI retrieval. Additionally, RetCCL enhances interpretability by highlighting diagnostic subregions within WSIs, providing insights into the search mechanism behind the WSI retrieval process. Specifically, in Stage A, new input features are integrated into the most semantically similar sub-memory banks, thereby improving the semantic consistency of stored features. This stage also utilizes weighted InfoNCE and group-level InfoNCE [165] to extract robust and universal features. Stage B involves the construction of an offline database for WSI retrieval and an online query process. While experimental results do not demonstrate a substantial performance improvement over its foundational model, MoCo v2 [137], the CCL-based feature extraction approach provides valuable insights into achieving stronger feature representations.

CAMIL [69] proposed a contrastively-clustered attention-based MIL model capable of directly predicting continuous biomarkers from WSIs. The experiment demonstrated that regression-based prediction scores offer higher prognostic value than classification-based scores. CAMIL consists of three main steps: (1) image preprocessing; (2) feature extraction; and (3) classification-based and regression-based attention for score aggregation, leading to patient-level predictions. During the preprocessing stage, CAMIL employs pre-trained ImageNet models and RetCCL [59] for feature extraction through SSL. In the subsequent stage, attention-based MIL processes the instance embeddings extracted in the previous step. The key innovation lies in the final step, where three independently trained regression heads are used. Extensive experiments confirmed that MIL achieves excellent performance in the prediction of continuous variables.

Ctranspath [58] marks a significant advancement in CPATH by leveraging SSL to reduce the reliance on manual annotations in CPATH tasks. It introduces a novel SSL strategy called semantically-relevant contrastive learning (SRCL), an enhanced framework based on MoCo v3 [166]. SRCL aligns multiple positive instances with similar visual concepts, thereby increasing the diversity of positive samples and yielding more informative representations. CtransPath employs a hybrid model that combines CNNs with a multi-scale Swin Transformer [167], enabling pretraining on large-scale unlabeled datasets. As a collaborative local–global feature extractor, it learns general feature representations that are better suited to histopathological image analysis tasks.

HIPT [148] proposed the Hierarchical Image Pyramid Transformer for the hierarchical aggregation of visual tokens and pretraining on gigapixel pathology images. HIPT is based on a core concept: visual tokens are fixed at a particular scale for a given magnification objective. The structure of HIPT consists of three components: cell-level, patch-level, and region-level representations, analogous to the hierarchical representations in natural language processing (e.g., character-, word-, sentence-, and paragraph-level representations). HIPT introduces two key innovations: (1) decomposing the task of learning robust WSI representations into hierarchically related sub-representations, each of which can be learned via SSL, and (2) leveraging DINO [168] to pretrain each aggregation layer with SSL. This hierarchical approach mirrors biological and medical phenomena, where layered relationships exist [169], such as DNA motifs influencing protein structure,

gene expression, and hereditary traits in exon sequences [170]. HIPT represents a significant step towards self-supervised slide-level representation learning by integrating mechanistic insights with advanced technical solutions.

UNI [78] is a general-purpose self-supervised model for pathology, pretrained on more than 100 million images. Although several SSL-pretrained image encoders for publicly available CPATH datasets have been proposed, UNI represents the first extensive development and evaluation of such methods across diverse tissue types. Based on ViT and DINO v2, UNI was pretrained on more than 100 million images from over 100,000 diagnostic H&E-stained WSIs (over 77 TB of data) spanning 20 major tissue types. It was subsequently evaluated on 34 tasks across multiple diagnostic domains. The results demonstrated UNI's superior performance in ROI-level classification, slide-level classification, segmentation, and disease subtype identification. By leveraging extensive data and experiments, UNI highlights the remarkable potential of SSL-based feature extractors in CPATH, establishing a robust foundation for data-efficient and generalizable AI models and pushing the boundaries of CPATH in both research and clinical applications.

ItS2CLR [67] introduced Iterative Self-paced Supervised Contrastive Learning for MIL Representations. To address performance degradation in traditional contrastive SSL (CSSL) methods due to data imbalance, ItS2CLR leverages instance-level pseudo labels derived from bag-level labels. This framework iteratively refines instance representations through supervised contrastive learning, guided by a self-paced learning (SPL) strategy to ensure the reliability of pseudo labels. Specifically, the framework begins with CSSL-based feature initialization, employs the DSMIL [61] aggregator to obtain instance-level probabilities, and alternates between refining the feature extractor, recalculating pseudo labels, and training the aggregator. This iterative process enhances the granularity of instance-level representations while improving both bag-level classification performance and the quality of pseudo-labels. ItS2CLR represents an optimization technique for feature extraction in MIL, effectively bridging the gap between bag-level supervision and instance-level representation learning, advancing feature extraction methodologies in CPATH.

4.2.2. Multimodal insights fusion

With the continuous advancement of MIL techniques, there is growing consensus among researchers that relying solely on the limited information from WSIs may not suffice to meet the diverse demands of downstream tasks. Consequently, fusing information from multiple modalities to achieve more comprehensive feature representations has become a prominent research focus in recent years. Two mainstream approaches have emerged for integrating data from other modalities: **modality alignment** and **modality aggregation**.

Modality alignment is exemplified by works such as CLIP [171]. The core idea of this approach is to use data from other modalities to guide the model in learning better image features, without requiring additional modality inputs during inference. In contrast, modality aggregation involves directly using other modalities as additional inputs to the model, extracting features across modalities to enhance representations.

Since the publication of CLIP in 2021, its modality alignment paradigm has gained significant attention, and earlier works in MIL have adopted this approach. Compared to images, text often encapsulates richer human prior knowledge. **PLIP** [149], which leverages a method similar to CLIP, extracted a large number of image–text pairs from Twitter to improve feature alignment, providing a valuable data collection strategy for subsequent studies. **MI-Zero** [64] extended CLIP-based methods to WSIs, enabling pretrained encoders to perform diverse downstream diagnostic tasks and significantly expanding the dataset for zero-shot transfer. Building on this, **CONCH** [95] further scaled the dataset to over 1.17 million image-caption pairs and introduced text generation capabilities alongside image–text alignment,

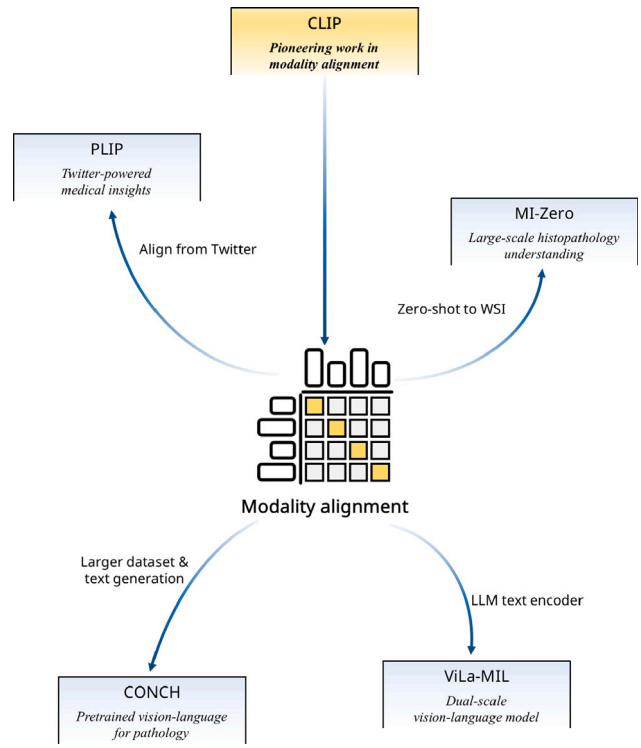

Fig. 9. Works involved in **modality alignment**.

optimizing image representation. **ViLa-MIL** [150] advanced these efforts by replacing the text encoder with a Large Language Model (LLM) embeddings, demonstrating superior feature representation and effectively integrating popular LLMs with MIL. Fig. 9 presents an overview of the works to be introduced below.

PLIP [149] introduced pathology language–image pretraining, a multimodal AI model capable of understanding both images and text for CPATH. PLIP addresses the critical issue of the limited availability of annotated medical images by curating and utilizing a large volume of de-identified images and knowledge shared by clinicians on public platforms, such as medical Twitter. Using popular pathology Twitter hashtags, PLIP compiled OpenPath, a dataset comprising 208,414 pathology image–text pairs. Although the model architecture aligns with Radford et al. [171], offering limited technical innovation, its novel data collection approach provides an effective solution for the annotation-scarce domain of CPATH, paving the way for further advancements.

MI-Zero [64] proposed a simple and intuitive MIL-based method that leverages the zero-shot transfer capability of pretrained visual language encoders for gigapixel-sized WSIs, which are routinely examined in clinical practice. A key innovation of MI-Zero lies in its ability to overcome the dual challenges of computational complexity and limited paired image–text datasets in CPATH. MI-Zero addresses these challenges by organizing a large dataset of pathology-specific image-caption pairs and utilizing contrastive alignment with image–text models, enabling the seamless integration of multimodal information. Furthermore, the model incorporates over 550,000 pathology reports and domain-specific text corpora to pretrain a robust text encoder, enhancing the synergy between visual and textual representations. The significance of this framework lies in its adaptability to real-world diagnostic tasks. By incorporating zero-shot transfer into the MIL paradigm, MI-Zero empowers pathologists to leverage pretrained encoders for downstream applications such as cancer subtyping and diagnosis. This multimodal fusion approach reduces reliance on labeled datasets, positioning MI-Zero as a pivotal work for scalable and

label-efficient solutions in CPATH.

CONCH [95] proposed the Contrastive Learning from Captions for Histopathology, a visual-language foundation model developed using over 1.17 million image-caption pairs through task-agnostic pretraining. CONCH aims to address the challenge of effectively integrating image and text data in pathology tasks. Traditional pathology models typically rely on image data, and their training processes are often constrained by the scarcity of annotated data, leading to poor generalization and limiting their applicability to specific tasks or diseases. To overcome these issues, CONCH is built on PMC-Path, a pathology-specific image-caption dataset derived from PubMed figures, forming a dataset that includes over 1.17 million human sample pairs. Based on the CoCa [172] framework, it employs an image encoder, text encoder, and multimodal fusion decoder to train the model using contrastive alignment objectives and captioning objectives.

CONCH has demonstrated exceptional performance across several tasks, including image classification, image segmentation, cross-modal image–text retrieval, and image caption generation. Additionally, in its specific workflow, CONCH provides detailed implementation methods for tasks such as processing figure captions, aligning subfigures with subtitles, and other similar tasks. CONCH provides a powerful visual-language pretraining model for the CPATH field, enabling effective integration of pathology images and text. This multimodal learning approach not only enhances the accuracy and generalization ability of pathology image analysis but also significantly expands the applicability of pathology tasks, making it suitable for handling data from different diseases and patient populations. Because CONCH undergoes task-agnostic pretraining, it can be directly applied to a variety of downstream tasks, reducing the need for task-specific supervised fine-tuning, and thereby offering broad practical value.

ViLa-MIL [150] proposed dual-scale vision-language MIL, which combines a vision-language model (VLM) with the MIL approach used for WSI prediction, making significant contributions to MIL feature extraction techniques. By leveraging the powerful feature extraction capabilities of VLM, ViLa-MIL addresses the issue of traditional MIL methods being susceptible to changes in data distribution. Additionally, it overcomes the problem of previous text prompts lacking consideration of pathological prior knowledge. The key components of ViLa-MIL include (1) a dual-scale visual descriptive text prompt; (2) a prototype-guided patch decoder; and (3) a context-guided text decoder.

The dual-scale visual descriptive text prompt captures both global structure and local details within the WSI by focusing on information at two scales. It uses a frozen LLM [173] to generate the text prompt. The prototype-guided patch decoder progressively aggregates patch features by grouping similar patches into the same prototype, allowing each prototype to capture more global contextual information for final similarity computation, thereby enhancing feature extraction efficiency through grouping and fusion. The context-guided text decoder incorporates multi-granular image contexts, including local patch features and global prototype features, using this contextual information to enhance the text features within the model.

ViLa-MIL successfully integrates the VLM, known for its outstanding feature extraction capabilities, with MIL, proposing a viable operational paradigm that has been validated through extensive experiments.

On the other hand, modality aggregation emerged later but follows a more intuitive approach: using multiple feature extractors to process distinct modalities and fusing the extracted features to achieve robust representations. **PIBD** [100], a non-application-focused study, employed a variant of the information bottleneck framework to address both intra- and inter-modality redundancy in multimodal data. **MMP** [101], building on PANTHER [174], established connections between optimal transport (OT) cross-alignment and Transformer cross-attention, optimizing the multimodal fusion process. **SurvPath** [99] advanced this approach by aggregating WSIs with transcriptomics, encoding cell-specific functional information to identify biological pathway markers. **HECTOR** [151] applied multimodal aggregation to achieve high-performance predictions of distant recurrence in endometrial cancer (EC). Fig. 10 provides an overview of the works introduced below.

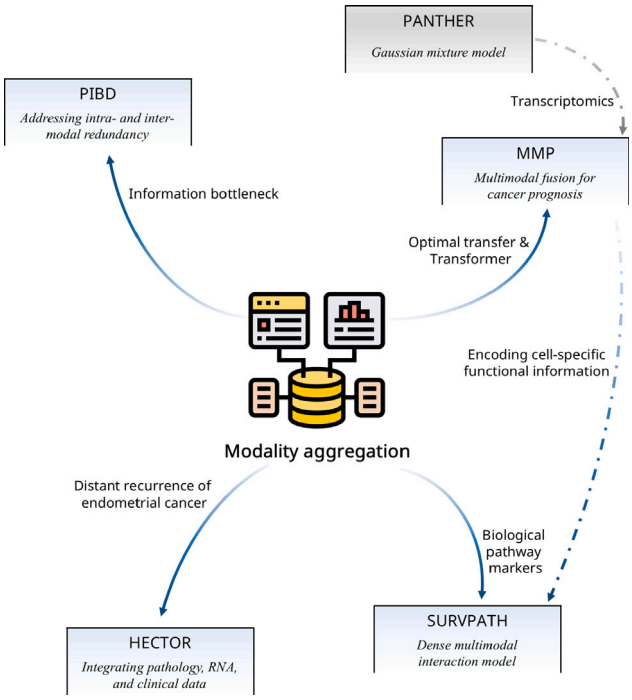

Fig. 10. Works involved in **modality aggregation**.

PIBD proposed Prototypical Information Bottlenecking and Disentangling to address the problem of excessive redundant information. It consists of two main components: the Prototypical Information Bottleneck (PIB) module, which handles intra-modal redundancy by selecting discriminative instances, and the Prototypical Information Disentanglement (PID) module, which separates multimodal data into modality-common and modality-specific knowledge. PIBD tackles a fundamental problem in multimodal MIL—redundant information—by decomposing complex multimodal information into easily distinguishable components, thus assisting the model in better feature extraction and instance classification.

MMP [101] introduced the multimodal Prototyping framework for patient prognostication to address the data scale issues in multimodal MIL. Compared to MIL approaches that rely solely on histopathology images for predictions, multimodal MIL integrates information such as WSIs and transcriptomic profiles, leading to a larger number of tokens. This increases memory requirements during training and adds complexity to interpretability analysis.

The main contributions of MMP are: (1) Morphological and Pathway Prototypes: MMP uses Gaussian Mixture Models (GMM) to construct compact WSI representations in an unsupervised manner. The mixture parameters define a slide summary that maps to 16 to 32 prototypes, achieving over 300-fold compression. Additionally, MMP transforms transcriptomics into a set of 50 Cancer Hallmark pathway prototypes [175]. By reducing the number of tokens, MMP allows direct processing of these compact multimodal tokens without relying on approximations. (2) Multimodal Fusion Mechanism: MMP establishes a connection between Optimal Transport (OT) cross-alignment and Transformer cross-attention, unifying both under a single framework. This not only reduces computational complexity but also improves the efficiency of the model, further optimizing the multimodal data fusion process.

SurvPath [99] integrated WSIs and bulk transcriptomics to predict patient survival, achieving state-of-the-art performance across five TCGA datasets. WSIs provide high-dimensional spatial tumor data,

while bulk transcriptomics capture internal gene expression. SurvPath addresses two key challenges: (1) Tokenizing transcriptomics: Semantically tokenizing transcriptomics data is challenging, as it differs from image or text tokenization. Direct fusion of feature vectors or grouping genes by function can lose semantic information. SurvPath overcomes this by tokenizing genes based on biological pathways [175–177]. (2) Capturing multimodal interactions: SurvPath introduces a novel attention mechanism to model complex interactions between images and transcriptomics, avoiding the computational expense of quadratic self-attention in Transformers. This mechanism focuses on patch-to-pathway, pathway-to-patch, and pathway-to-pathway interactions, while approximating patch-to-patch relationships.

The SurvPath framework includes three main components: (1) the Pathway Tokenizer, which encodes transcriptomics data into pathway embeddings; (2) the Histology Patch Tokenizer, which uses a Swin Transformer to convert WSI patches into embeddings [58,178]; and (3) multimodal fusion, where the attention mechanism reduces complexity by focusing on key interactions. SurvPath represents a significant advancement in multimodal MIL, improving transcriptomics interpretability and offering breakthroughs in modality fusion. Extensive experiments show its superior performance.

HECTOR [151] proposed Histopathology-based Endometrial Cancer Tailored Outcome Risk, a multimodal MIL model, to apply MIL methods to the prediction of distant recurrence in endometrial cancer (EC) and address the costly nature of the gold standard involving combined pathological and molecular profiling. HECTOR integrates anatomical stage information [179], instance embeddings obtained via ABMIL, and molecular subtypes derived from the im4MEC DL model through a gating-based attention mechanism [180,181]. Finally, the risk score is computed through a fully connected layer. While HECTOR does not strictly follow the conventional two-step MIL process, it leverages MIL's feature extraction and aggregation techniques using only bag-level labels. Although it does not propose a highly innovative algorithm, it provides a viable paradigm for multimodal MIL, with comprehensive experiments demonstrating its effectiveness.

4.3. Summary

This section underscores the significant advancements in MIL research for CPATH, driven by innovations in architecture design and feature extraction methods. The exploration of MIL architectures reflects the field's growing sophistication in optimizing instance selection and contextual relationship modeling, enabling more efficient and effective analysis of large-scale pathology data. Simultaneously, advancements in feature representation, such as self-supervised learning and multimodal integration, have played a crucial role in addressing data scarcity and enhancing feature representation quality [182]. Together, these developments illustrate the continuous evolution of MIL as a foundational technology in computational pathology, paving the way for more accurate, scalable, and innovative diagnostic solutions.

The first trajectory focuses on refining MIL architectures to improve performance in pathology tasks. Two central themes emerge within this domain: Valuable Instance Selection and Inter-instance Relationship Fusion. Valuable Instance Selection addresses the challenge of managing the massive scale of WSIs by identifying and prioritizing the most relevant regions or instances. This approach significantly reduces computational demands while ensuring that the most informative data is utilized for downstream tasks. Over time, techniques in this area have progressed from straightforward selection strategies to more sophisticated methods that enhance selection accuracy and efficiency. Inter-instance Relationship Fusion shifts the emphasis from individual regions to the contextual relationships among instances within WSIs. By leveraging architectural innovations, this approach captures richer contextual information, enabling models to better understand the complex spatial and relational structures inherent in pathology data. Recent advancements in this domain have markedly improved computational efficiency and scalability, facilitating the processing of larger and more intricate datasets.

The second trajectory emphasizes advancing feature extraction techniques to fully realize MIL's potential in CPATH. This trajectory is divided into two key areas: Self-Supervised Discoveries and Multimodal Insights Fusion. Self-Supervised Discoveries leverage the capacity to learn from unlabeled data, addressing the challenges posed by limited annotations in CPATH. By integrating self-supervised learning frameworks, these methods extract more robust and generalized feature representations, leading to improved performance across a diverse array of tasks. Multimodal Insights Fusion expands the scope of data utilization by incorporating information beyond WSIs, such as pathology reports or molecular data. Two primary strategies have emerged: aligning features across modalities to guide superior image representation and aggregating multimodal inputs to create enriched and comprehensive feature sets. These approaches mitigate the limitations of unimodal analysis and provide holistic insights for downstream tasks.

5. Discussion

In Sections 3 and 4, we reviewed recent trends and key innovations in MIL technology for CPATH. Despite the emergence of various clinical applications and technical advancements in recent publications, the core development of MIL methods can fundamentally be divided into two components: feature extraction and feature aggregation. The former focuses on extracting fine-grained, low-dimensional embeddings, while the latter uses these embeddings to generate a robust WSI-level representation. Whether through architectural improvements or advancements in feature extraction, the ultimate goal of MIL remains to produce a reliable and representative WSI-level representation.

Several areas warrant further exploration, including interpretability, post-inference adaptability to downstream tasks, and clinical applications [183–185]. The core structure of MIL, along with its various stages throughout the entire workflow, defines the primary research directions for MIL in CPATH at this stage. This section highlights key aspects deserving future research, providing analysis and informed predictions.

5.1. From unimodal to multimodal

In CPATH, WSIs have long been central to feature extraction in MIL, evolving from traditional feature engineering and CNNs to ViT-based models capable of capturing global and cross-scale features [186]. However, the reliance on WSIs alone presents inherent limitations. While WSIs provide crucial insights into cellular and tissue structures, they often fall short in addressing the complexity of many pathology tasks, such as fine-grained cancer subtyping and prognosis prediction. These tasks require a deeper understanding of disease mechanisms that WSIs alone cannot offer, thus necessitating the integration of additional data modalities [187].

The primary limitations of unimodal methods are twofold: (1) WSIs often lack sufficient information to predict complex pathological outcomes, and (2) WSI-based pretraining primarily captures image features without the depth offered by other modalities, such as genomic or clinical data. Multimodal learning addresses these gaps by combining diverse data sources to create a more comprehensive representation of disease processes, thereby enhancing the model's predictive power and generalization capabilities [188].

Recent advancements have focused on multimodal feature extraction and fusion. Effective multimodal fusion requires balancing the contributions from each modality while mitigating issues such as redundancy and conflicts [189]. Moreover, real-world medical scenarios often involve missing modalities, making robustness to incomplete data a critical area of ongoing research [189]. These developments aim to enable models to seamlessly integrate image, genomic, and clinical data, fostering a holistic understanding of disease mechanisms.

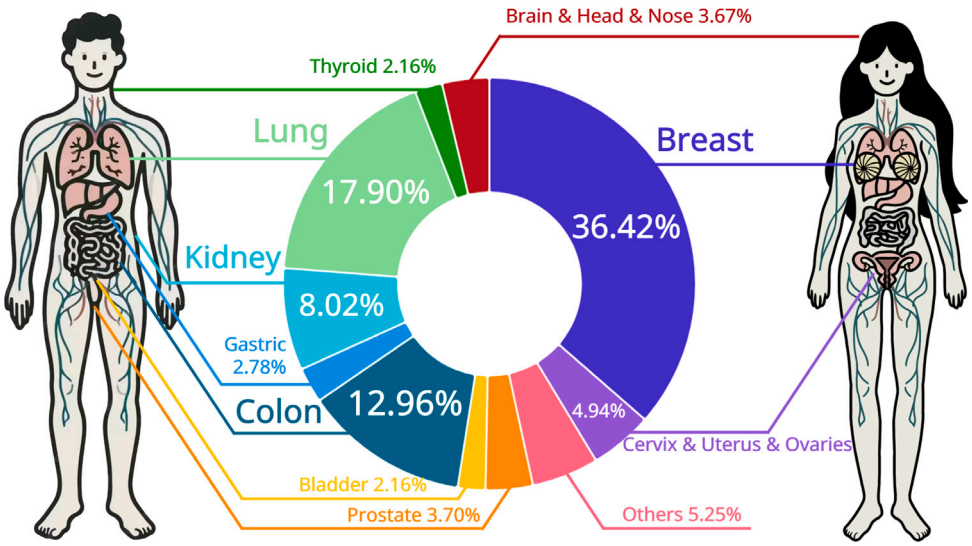

Fig. 11. Distribution of anatomical locations in related works.

Multimodal pretraining is an especially promising approach [190]. By leveraging data from diverse modalities, pretraining enables models to learn deeper and more meaningful features from WSIs. This strategy enhances generalization and provides a foundation for tackling more complex pathology challenges.

As multimodal data becomes increasingly accessible, robust multimodal models will play an increasingly critical role in pathology. By combining high-quality feature extraction with advanced information fusion techniques [191], these models promise to improve personalized clinical decision-making and deepen our understanding of intricate disease mechanisms [192]. The future of CPATH lies in harnessing the synergy between modalities to overcome the limitations of unimodal approaches [193].

5.2. Integrating LLMs

The integration of large language models (LLMs) with MIL represents a transformative direction for CPATH [191]. LLMs, with their advanced language comprehension and generation capabilities [194], can bridge the gap between complex textual data (e.g., clinical notes, pathology reports) and visual information from WSIs, creating a more comprehensive framework for pathology analysis [27]. By combining WSI analysis with LLMs, researchers aim to unlock new possibilities in feature extraction, interpretability, and task-specific adaptability.

Firstly, WSI analysis often struggles to fully encompass the complexity inherent in numerous pathological tasks. While WSIs provide rich visual information about cellular and tissue structures, they lack the contextual and biological insights offered by other data types, such as pathology reports or genomic information. By integrating LLMs, the MIL model can effectively utilize complementary modalities, enhancing feature extraction through textual domain expertise [195]. This text-driven approach improves both the adaptability of MIL models and their ability to address a variety of downstream tasks.

Secondly, the application of LLMs in data annotation and few-shot learning addresses the challenge of limited labeled data in pathology. Annotating WSIs is both labor-intensive and costly, but LLMs can generate high-quality pseudo-labels [196], enabling MIL models to train efficiently even with sparse annotations. Moreover, in few-shot learning scenarios, LLMs can provide rich semantic descriptions of small datasets, adding contextual depth to the training process [197]. This dual approach—combining model-driven learning with data-driven insights—paves the way for more robust and scalable MIL models capable of handling complex pathological tasks.

Thirdly, LLMs significantly enhance the capabilities of multimodal MIL models by enabling zero-shot and class-agnostic task handling. Unlike traditional approaches that require dedicated classification heads for each task, LLMs utilize prompts to guide the model in addressing a wide range of tasks without task-specific modifications. For instance, in disease subtyping or treatment recommendations, LLMs can seamlessly incorporate unstructured textual data, enriching WSI feature embeddings with contextual information from clinical narratives. This flexibility allows the model to adapt more effectively to diverse clinical scenarios than traditional multimodal methods, eliminating the need for task-specific engineering.

Finally, the integration of LLMs enhances the interpretability of MIL models, a critical factor in clinical applications. LLMs can generate natural language explanations for model predictions, offering clinicians intuitive and transparent insights [198]. For example, in classifying a tumor subtype, an LLM-powered MIL model could provide a detailed rationale by linking visual patterns in WSIs with relevant textual evidence. This level of explainability fosters trust in AI systems and supports their adoption in real-world clinical settings [199].

By integrating LLMs with WSI-based analysis, future research can overcome the limitations of unimodal approaches, delivering more accurate, interpretable, and generalizable models for CPATH. This synergy between vision and language represents a critical step towards advancing AI in clinical pathology.

5.3. Rare pathologies

The distribution of specific diseases addressed in MIL studies over the past six years was analyzed, with the results displayed in Fig. 11. The analysis reveals that research on breast cancer accounts for 36.42% of all studies, making it the most thoroughly examined disease. This prominence is largely attributable to the highly-regarded CAMELYON16 and CAMELYON17 datasets, which have greatly influenced the advancement of MIL in CPATH. Following breast cancer in study frequency are lung cancer, colon cancer, and kidney cancer, supported by robust datasets like TCGA, and these cancers also possess significant clinical importance.

The proportional distribution highlights that the majority of MIL research has concentrated on a few key cancer types, reflecting the heavy reliance on public datasets in current studies. Studies focusing on all other cancer types account for only 24.7%, highlighting the significant disparity in attention given to different cancer types within CPATH. This discrepancy is primarily due to the varying levels of

database completeness. Some rarer cancers still lack comprehensive datasets, making it challenging to provide sufficient data for model training. Applying MIL to rare pathologies presents a significant challenge due to the limited availability of data samples, which hinders the effective training of robust, high-performance models using traditional deep learning methods [200].

One potential solution is domain generalization [201], which involves learning generalized feature representations from data-rich domains and applying them to rare pathologies to compensate for the lack of direct training data. This can be achieved through shared feature spaces [202] or transfer learning [203], enabling the model to maintain strong performance in rare pathologies. Furthermore, MIL models must focus on effectively learning from limited labeled samples. Few-shot learning [204] may become a crucial research direction for rare pathologies. By incorporating few-shot learning, MIL models can extract key features related to pathological characteristics even with scarce training data. These methods might include contrastive learning, SSL, or generating pseudo-labels to enhance the model's ability to recognize rare pathologies [205].

5.4. Key challenges in clinical applications

The successful integration of MIL into clinical practice faces several key challenges, including interpretability, workflow integration, and privacy concerns. Each of these aspects is critical to ensuring that MIL models can be trusted and widely adopted in real-world medical environments.

Interpretability of MIL is paramount in the medical field, where clinicians need to trust and understand the decision-making process of deep learning models [206,207]. Although MIL models have demonstrated remarkable performance in classification and prediction tasks, their utility in clinical settings hinges on providing transparent and interpretable results [208]. Given the inherent nature of machine learning, biased predictions are not uncommon in classifiers [209,210]. Thus, clinicians must be able to comprehend the rationale behind the model's predictions, not merely receive a classification output [211,212]. Current instance-based methods and attention mechanisms offer some level of interpretability, particularly by highlighting critical regions in whole-slide images (WSIs). However, these explanations are often limited to visual cues, and there is significant room for improvement in providing more granular, actionable insights. Future MIL models should aim to offer multimodal explanations, including natural language descriptions that clearly articulate the reasoning behind their decisions [213]. This approach would align the model's output with the cognitive processes of clinicians, making it more useful in practice.

In addition to interpretability, embedding MIL into diagnostic workflows presents another crucial challenge for the practical application of CPATH [214,215]. While MIL models excel in quantitative predictions, such as disease subtyping or positive/negative classifications, pathologists require more than just a final decision. They need diagnostic support functionalities, such as the annotation of key lesion areas, visualization of features, and interpretative explanations of different pathological patterns. To meet clinical operational needs, future MIL models must provide multi-level outputs that extend beyond mere predictions and offer supplementary diagnostic information. Moreover, these models need to demonstrate robustness and reliability across various hospital and laboratory conditions, where scanning methods, staining techniques, and other factors may vary significantly [216]. This necessitates models with strong generalization capabilities that can perform consistently in diverse, real-world environments [105].

Privacy in MIL is also emerging as a critical research direction as MIL models become more widely adopted in clinical settings [217,218]. Pathological images and their associated clinical data often contain sensitive personal health information, raising significant privacy concerns. Two key areas of focus in addressing privacy challenges are the integration of federated learning with MIL for collaborative research across institutions and the application of encrypted computation techniques. Federated learning allows MIL models to be trained across multiple institutions without requiring the sharing of sensitive data [219]. Meanwhile, homomorphic encryption techniques enable models to perform computations on encrypted data, ensuring that the original data remains protected during both training and inference processes [220,221]. Given the sensitive nature of healthcare data, addressing privacy concerns will be essential for the widespread adoption of MIL in CPATH.

6. Conclusion

In recent years, CPATH has witnessed a substantial surge in the application of MIL techniques, leading to transformative advancements in the field. This review addresses a critical gap in the literature by providing a detailed and systematic examination of MIL developments over the past six years, overcoming the limitations of prior works that lacked comprehensive coverage or a clear technological roadmap. Specifically, this review: (1) presents a structured and generalized framework for MIL in CPATH, clarifying its foundational principles; (2) meticulously compiles, analyzes, and synthesizes key studies, uncovering major trends, technical advancements, and pivotal turning points in the evolution of MIL; (3) highlights innovative approaches in MIL architecture design and feature representation, which have significantly enhanced the capacity to handle complex tasks; and (4) offers forward-looking insights into future research directions and emerging methodologies.

Our findings highlight the remarkable progress driven by innovations in feature extraction and integration, which have enhanced MIL's ability to deliver robust and precise representations. These advancements have facilitated more effective support for increasingly complex downstream tasks and expanded the range of application scenarios in CPATH. Furthermore, the minimal reliance on exhaustive annotations and the inherent interpretability of MIL frameworks underscore their immense potential to address the challenges of clinical pathology data and advance diagnostic and prognostic applications.

Looking ahead, the trajectory of MIL in CPATH suggests a growing emphasis on cross-modal integration and advanced feature extraction techniques. As the field continues to evolve, MIL is uniquely positioned to drive breakthroughs that bridge computational innovation and clinical impact. We hope this review serves as a comprehensive resource for researchers and practitioners, guiding future exploration and fostering the continued development of MIL in CPATH.

CRediT authorship contribution statement

Yuchen Zhang: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Conceptualization. **Zeyu Gao:** Writing – review & editing, Validation, Supervision, Investigation. **Kai He:** Writing – review & editing, Validation, Supervision, Investigation. **Chen Li:** Supervision, Resources, Project administration, Funding acquisition, Conceptualization. **Rui Mao:** Writing – review & editing, Validation, Resources, Investigation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

This work has been supported by the Key Research and Development Program of Ningxia Hui Nationality Autonomous Region under Grant 2022BEG-02025 and 2023BEG-02023.

Data availability

No data was used for the research described in the article.

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